Initial results towards a new 'benchmark' assessment of the South African anchovy resource

C.L. de Moor*

Correspondence email: *carryn.demoor@uct.ac.za*

A 'benchmark' assessment of the South African anchovy resource is currently underway which will provide Operating Models for the upcoming Management Strategy Evaluation to simulation test a new joint Operational Management Procedure for sardine and anchovy. This document presents initial results, comparing between key model assumptions in order to select a baseline model. These include assumptions regarding natural mortality, stock-recruitment relationships and time-varying growth and commercial selectivity curves.

Keywords: anchovy, assessment, natural mortality, population dynamics, South Africa, stock recruitment relationships

Introduction

A 'benchmark' assessment of the South African anchovy resource is currently underway. This benchmark involves not only updating the data used in the assessment, but also reconsiders model assumptions. This document presents initial results, comparing between key model assumptions in order to select a baseline model.

Methods

South African anchovy are assumed to form a single stock distributed along the west and south coasts of South Africa. The integrated assessment is age structured, fitting to length structured data. The population dynamics model is detailed in Appendix A and the survey and commercial data available for this assessment from 1984 to 2023 have been detailed in de Moor *et al*. (2024). The model parameters are defined in Tables A.1 and A.2. Key methodological changes since the last assessment by de Moor (2020a,b) include:

i) Model quarters are now January-March, April-Jun, July-September and October-December, instead of November – January, February – April, May – July, August - October.

ii) Spawning and recruitment are assumed to occur at the start of the 4th quarter, 1st October, instead of 1st November. October has been a peak month for anchovy spawning out of an October-January spawning season as observed in 1977-78 the Cape Egg and Larval Program (CELP) and 1993/4-1994/5 SARP surveys (Painting and Korrûbel 1998; van der Lingen and Huggett 2003), although December was found to be the peak month in ichthyoplankton surveys in the 1960s and Sardine and Anchovy Recruitment Programme (SARP) line from 1995-2001 (Mullon *et al*. 2003).

iii) The November hydro-acoustic survey is assumed to provide an estimate of relative abundance corresponding to the total biomass at 15th November (Figure A.1 of de Moor 2022), rather than 1st November.

Some key alternative assumptions are tested in this document to guide the selection of a baseline model for this benchmark assessment. These include (i) alternative stock recruitment relationships, (ii) alternative rates of natural mortality, and (iii) whether to allow for early/late recruitment.

<ins>Stock recruitment relationship (with $\bar{M}_j = \bar{M}_{ad} = 1.2$ and $t_{0,y} = t_0$)</ins>

The following alternative stock recruitment relationships have been considered (Table 1):

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* MARAM (Marine Resource Assessment and Management Group), Department of Mathematics and Applied Mathematics, University of Cape Town, Rondebosch, 7701, South Africa.

A_{BH} - Beverton Holt stock-recruitment curve, with uniform priors on steepness and carrying capacity

A_{2BH} - two Beverton Holt stock-recruitment curves, with uniform priors on steepness and carrying capacity, one estimated using data from 1984 to 1998 and the other from 1999 to 2023

A_{3BH} - two Beverton Holt stock-recruitment curves, with uniform priors on steepness and carrying capacity, one estimated using data from 1984 to 1998 and 2013 to 2023, and the other from 1999 to 2012

A_{R} - Ricker stock-recruitment curve, with uniform priors on steepness and carrying capacity

A_{ModR} - Modified Ricker stock-recruitment curve, with uniform priors on steepness and carrying capacity

A_{HS} - hockey stick stock-recruitment curve, with uniform priors on the log of the maximum recruitment and on the ratio of the spawning biomass at the inflection point to carrying capacity

A_{2HS} - two hockey stick stock-recruitment curves, with uniform priors on the log of the maximum recruitment and on the ratio of the spawning biomass at the inflection point to carrying capacity, one estimated using data from 1984 to 1998 and the other from 1999 to 2023.

In cases where a second curve is estimated from 2000 to 2012 or 2023, the variance about the stock recruitment curve over this time period, $\sigma^2_{r,2000+}$, is estimated separately from that for the earlier time period, σ^2_r.

Natural mortality (with A_{BH} and $t_{0,y} = t_0$)

A number of combinations of time-invariant median juvenile and adult natural mortality values have been explored, covering the range 0.6 to 1.8 year^{-1}. For realism, only combinations with $\bar{M}_j > \bar{M}_{ad}$ have been considered. Alternatives to the assumption of time-invariant natural mortality were considered through the following robustness tests:

A_{Mad} – annually varying adult natural mortality, i.e. random effects model with $\sigma_{ad} = 0.2$ or $\sigma_{ad} = 0.1$, and $\rho \sim U(0,1)$.

A_{Mj} – annually varying juvenile natural mortality, i.e. random effects model with $\sigma_j = 0.2$ or $\sigma_j = 0.1$, and $\rho \sim U(0,1)$.

Age at which the length is zero for growth curve (with $\bar{M}_j = \bar{M}_{ad} = 1.2$ and A_{BH})

This alternative tests the impact of allowing for inter-annual variability in the timing of peak spawning, i.e. to allow for early/late recruitment:

A_{t0} - annually varying t_0

In this alternative, equations A.6 and A.7 are adjusted as follows:

$$A^{Oct}_{y,a,l} \sim N\left(L_\infty\left(1+(D-1)e^{-\kappa(a-t_{0,y-a})}\right)^{1/(1-D)}, \vartheta_a^2\right) \qquad 0 \le a \le 4^+, 1.5^- cm \le l \le 16^+ cm$$

$$A_{y,q,a,l} \sim \begin{cases} N\left(L_\infty\left(1+(D-1)e^{-\kappa(a+(2q+1)/8-t_{0,y-a})}\right)^{1/(1-D)}, \left(1-\frac{2q+1}{8}\right)\vartheta_a^2 + \frac{2q+1}{8}\vartheta_{a+1}^2\right) & 1 \le q \le 3 \\ N\left(L_\infty\left(1+(D-1)e^{-\kappa(a+1/8-t_{0,y-1-a})}\right)^{1/(1-D)}, \left(1-\frac{1}{8}\right)\vartheta_a^2 + \frac{1}{8}\vartheta_{a+1}^2\right) & q = 4 \end{cases}$$

$$0 \le a \le 4^+, 1.5^- cm \le l \le 16^+ cm$$

with $t_{0,y} = t_0 + \varepsilon^t_y$

and $\varepsilon^t_y = \begin{cases} \eta^t_y & y = y_0 \\ \rho^t \varepsilon^t_{y-1} + \sqrt{1-(\rho^t)^2}\,\eta^t_y & y_1 < y \le y_n \end{cases}$

t_0 - average age at which the length (in expectation) is zero ($t_0 = \frac{1}{\kappa_c}\left\{ln\left(\frac{L_{c,1}-L_{c,\infty}}{L_{c,\infty}}\right) + \kappa_c\right\}$, where $L_{c,a=1} \sim U(5,18)$)

ε^t_y - annual autocorrelated residuals about the age at which length is zero

$\eta^t_y \sim N(0, 0.2^2)$ - annual uncorrelated residuals about the age at which length is zero

$\rho^t \sim U(-1,1)$ - autocorrelation coefficient in these residuals

Commercial selectivity (with $\overline{M}_j = \overline{M}_{ad} = 1.2$ and A_{BH})

A greater proportion of anchovy catch has been taken south/east of Cape Point since ~2003 (Coetzee *et al*. 2024). As larger/older fish are found further south/east, this test allows for a change in commercial selectivity in 2003/2004:

A_{sel} - commercial selectivity-at-length is estimated separately for 1984-2003 and 2004-2023.

Results and Discussion

Results are given at the joint posterior mode only. A subsequent document will show the full posterior distributions. Figures 1 to 12 show the model fit to the data and associated key estimated relationships with $\overline{M}_j = \overline{M}_{ad} = 1.2$, A_{BH}, and $t_{0,y} = t_0$. The model fit to the hydroacoustic survey data and DEPM estimates is relatively good (Figures 1 to 3). There are three years (1996, 2011 and 2019) when the model does not predict a biomass as low as that observed; however relatively high recruitments were observed just preceding these years in 1995, 2007-2010, 2018-2019. While the model tracks the general trend of observed recruitments, the fit to the recruit timeseries is not as good. This is not purely an indication of a changing year-long juvenile natural mortality rate (see A_{t0} below).

The model fits to the proportion-at-length data, with time-invariant survey and commercial selectivity-at-length relationships are reasonable (Figures 5, 7); the fixed points in the commercial selectivity-at-length relationships were selected after considering a model with commercial selectivity-at-length relationships varying for each quarter (Appendix C; the baseline -lnPosterior is -793.95, compared to -795.02 with the additional 5 parameters). The residuals in Figure 7 do not indicate any obvious need to allow for a change in selectivity over time (years; see A_{sur} below).

While the model structure estimates recruitment at 1st October for consistency reasons, as both sardine and anchovy models will be combined in the Operating Model for the next Management Strategy Evaluation, the effective timing of peak recruitment as estimated by the model is approximately early-December (Figure 8, $t_0 = 0.20$). This corresponds with approximately the middle of the October-January spawning season (Painting and Korrûbel 1998; Mullon *et al*. 2003; van der Lingen and Huggett 2003).

The estimated Beverton-Holt stock recruitment relationship is shown in Figure 10 and the timeseries of November recruitments in Figure 11 (noting recruitment in November 2023 may not be well estimated given the data available).

Natural mortality

Table 2 lists the various contributions to the negative log posterior probability distribution function (pdf) at the joint posterior mode for the full range of combinations of juvenile and adult natural mortality explored. There is a little change in the posterior distribution as $\overline{M}_j$ is changed for a given $\overline{M}_{ad}$. However, given $\overline{M}_j$, the posterior distribution indicated an improved fit to the data for increasing $\overline{M}_{ad}$. This latter feature has been observed in previous assessments and may be an artefact of the assessment methodology in that a higher natural mortality results in a higher loss of "memory" of cohorts, making the November survey data easier to fit. Alternatives which result in $k_r/k_N \leq 0.5$ are considered unrealistic. The data thus support the continued use of $\overline{M}_j = \overline{M}_{ad} = 1.2$ as the preferred time-invariant alternative.

Figure 12 shows the annual estimated juvenile and adult natural mortality rates when these are (separately) allowed to vary interannually (A_{Mj} and A_{Mad}). A decrease in juvenile natural mortality over the turn of the century, corresponding to the pulse in anchovy recruitment and biomass results in an improved fit to the data. In addition, there may be some indication of a small increase in juvenile natural mortality in the most recent decade, while the 1980-90s experienced greater changes in juvenile natural mortality. In contrast, there is a trend in the estimated adult natural mortality over time. While the annual estimates of juvenile natural mortality are similar to those estimated by de Moor (2020b), the trend in estimated adult natural mortality was not as evident with data up to 2019 (de Moor 2020b). While one needs to be cautious that these models which fit the data better do not primarily reflect a 'fitting to noise' (Smith *et al*. 2011), they remain informative, particularly from an ecosystem perspective.

These initial results suggest there is support for allowing some change in natural mortality over time in the baseline model(s). Alternatives such as A_{Mj} and A_{Mad} are preferred to options previously considered (see e.g. de Moor 2020b) which assumed an estimated increase in natural mortality at a pre-determined year and which was then assumed to continue indefinitely. The juvenile natural mortality rates have a slightly lower estimated autocorrelation coefficient ($\rho = 0.64 \left(\sigma_j = 0.1\right), 0.53 \left(\sigma_j = 0.2\right)$), compared to that estimated for the adult natural mortality rates ($\rho = 0.68\ (\sigma_{ad} = 0.1), 0.71\ (\sigma_{ad} = 0.2)$). Another proposed alternative which has not yet been tested is to allow for density dependent natural mortality to account for the greater relative predation on forage fish species when their biomass is low. Saraux *et al*. (2021) showed the % of anchovy stock consumed by Gannets increased at low anchovy biomass. Their analysis used the survey estimates of biomass west of Cape Agulhas only. de Moor (2020b) tested a simple density dependent option of $\bar{M}_{j,y} = \bar{M}_{ad,y} = \bar{M} + e^{-\chi B_{y-1}}$, but the data supported a relatively flat relationship between natural mortality and November biomass. Alternative relationships that allow for increases in natural mortality at both high and low stock size and/or the possibility of density dependence operating on either juvenile or adult natural mortality could be investigated. Note, however, these initial results all assume a time-invariant Beverton Holt stock recruitment relationship.

<u>Stock recruitment relationship</u>

Table 3 lists the various contributions to the negative log posterior probability distribution function (pdf) at the joint posterior mode for the alternative stock recruitment relationships, while Figure 13 shows the estimated relationships. Although overall improvements to the posterior are achieved when further parameters and additional relationships are estimated (e.g. A_{2BH}, A_{3BH}, A_{2HS}), this improvement is primarily achieved through an improvement in the contribution from the prior distribution for the recruitment deviates, and only a small improvement from the likelihood (the fit to the data). This is similar to that found by de Moor (2020b). Furthermore, when more than one relationship is estimated, the model prefers (almost) no (or an (almost) flat) relationship during at least one period of time. This differs from that estimated by de Moor (2020b) where a flat relationship was only estimated for the most recent years of the time series.

The time-invariant hockey-stick relationship provides a slightly better fits to the data than that for the Beverton Holt stock recruitment relationship (Table 3). However, the different relationships could have substantially different implications for the risk to the resource in MSE simulations and thus choice(s) should be made carefully as to which form the baseline model and which are part of the robustness tests.

<u>Variability in timing of peak recruitment</u>

A substantially better fit to the survey proportion-at-length data is obtained when allowing for inter-annual variability in the timing of peak spawning (A_{t0}) (Table 3), although this is not noticeable in the average (over all years) fit (Figure 5). While the t_0's do vary by more than a month over time and recruitment is estimated, on average, to be later (Figure 8), there is no substantial trend in this variability. Thus, this does not lend support for concerns that there may be a substantial shift in peak recruitment over time towards later in the year as seen elsewhere (e.g. Erauskin-Extramiana *et al*. 2019) and considered due to the decline in average caudal length of anchovy landed in Pool Areas B and C (DFFE 2020).

Commercial selectivity

Allowing for a change in commercial selectivity between 2003 and 2004 results in an improved fit to the data (Table 3), although there is only a (small) change to selecting fewer smaller fish in quarters 2 to 4 from 2004- in the estimated quarterly commercial selectivity-at-length relationships between the two time-periods (Figure 6). Increasing the quarterly break-points to allow for maximum selectivity of larger fish results in a worse fit to the data (results not shown), and thus the same break-points were used for both A_{BH} and A_{sel}.

Summary and Next Steps

Some key alternative assumptions have been tested in this document to guide the selection of a baseline model for this benchmark assessment of South African anchovy. These include (i) alternative stock recruitment relationships, (ii) alternative rates of natural mortality, (iii) whether to allow for inter-annual variability in the timing of recruitment, and (iv) whether to allow for a change in commercial selectivity over time. All comparisons were done while holding other assumptions constant (i.e. assuming a Beverton-Holt stock recruitment relationship, a time-invariant natural mortality rate of 1.2year$_{-1}$ and/or a time-invariant growth curve). These results indicate that assuming a time-invariant growth curve will suffice for a baseline model for anchovy, while time-varying natural mortality might need to be accounted for in the baseline model. Time invariant stock-recruitment relationships were preferred while assuming time-invariant M, but this will need to be re-confirmed if the baseline model assumes time-varying natural mortality.

Once a baseline model is selected, further results such as retrospective runs, historical harvest proportions and comparisons with dynamic B_0 will be produced. A set of robustness tests should also be selected; if the model appears insensitive to some of these they may not be used to test the robustness of the next OMP. The set of robustness tests is currently proposed to include the following:

A_M - alternative fixed and/or time-varying and/or density dependent natural mortality options

A_{SR} - alternative stock recruitment relationships

A_4 - No plus group, all remaining fish assumed to die as they reach age 5.

A_{sur} - Survey selectivity below 7.5[1]cm was estimated to be a constant, and uniform (1) selectivity was assumed for lengths ≥7.5cm.

A_{com} - Commercial selectivity was not estimated to decrease at higher lengths, i.e. $\delta_q = 0$.

A_{com2} - Commercial selectivity was modelled using a double-logistic curve.

A_{kegg} - Egg survey bias estimated with uninformative prior, i.e. $ln\left(k_g^A\right) \sim U(-100{,}0.7)$.

[1] This length was chosen as it provided the best deterministic fit in the range 6-9cm.

A_{kegg1} - Negatively biased egg surveys, i.e., $k_g^A = 0.75$.

A_{kegg2} - Positively biased egg surveys, i.e., $k_g^A = 1.25$.

A_{lamR} - Additional variance (over and above the survey sampling CV) associated with the recruit survey fixed $(\lambda_r^A)^2$=0.

A_{lamN} - Additional variance (over and above the survey sampling CV) associated with the November survey, estimated with $(\lambda_N^A)^2 \sim U(0,100)$.

A_{lamN2} - Additional variance (over and above the survey sampling CV) associated with the November survey fixed $(\lambda_N^A)^2 = 0.02$.

A_{growth} - Richards growth curve rather than von Bertalanffy growth curve.

A_{like} - Multinomial likelihood for proportion at length data.

A_{mat} - Time-invariant maturity ogive.

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Table 1. The alternative stock-recruitment relationships considered. The parameters are defined in Appendix A, Table A.1, with, $X = \sum_{a=1}^{3} \bar{f}_a w_a e^{-\bar{M}_j-(a-1)\bar{M}_{ad}} + \bar{f}_{4+} w_{4+} e^{-\bar{M}_j-3\bar{M}_{ad}} / \left(1 - e^{-\bar{M}_{ad}}\right)$.

Test	Stock recruitment relationship	$f(SSB_y)$	Parameters
A_{BH}	Beverton Holt	$\frac{\alpha SSB_y}{\beta + SSB_y}$	$h \sim U(0.2,1)$ $K/1000 \sim U(0,10)$ $\alpha = \frac{4h}{5h-1}\frac{K}{X}$ $\beta = \frac{K(1-h)}{5h-1}$
A_{2BH}	Beverton Holt (2 curves)	$\begin{cases} \frac{\alpha_1 SSB_y}{\beta_1 + SSB_y} & \text{if } y < 1999 \\ \frac{\alpha_2 SSB_y}{\beta_2 + SSB_y} & \text{if } y \geq 1999 \end{cases}$	$h_t \sim U(0.2,1)$ $K_t/1000 \sim U(0,10)$ $\alpha_t = \frac{4h_t}{5h_t-1}\frac{K_t}{X}$ $\beta_t = \frac{K_t(1-h_t)}{5h_t-1}$ $t = 1,2$
A_{3BH}	Beverton Holt (2 curves, 3 periods)	$\begin{cases} \frac{\alpha_1 SSB_y}{\beta_1 + SSB_y} & \text{if } y < 1999, y > 2012 \\ \frac{\alpha_2 SSB_y}{\beta_2 + SSB_y} & \text{if } 1999 \leq y \geq 2012 \end{cases}$	$h_t \sim U(0.2,1)$ $K_t/1000 \sim U(0,10)$ $\alpha_t = \frac{4h_t}{5h_t-1}\frac{K_t}{X}$ $\beta_t = \frac{K_t(1-h_t)}{5h_t-1}$ $t = 1,2$
A_R	Ricker	$\alpha SSB_y e^{-\beta SSB_y}$	$h \sim U(0.2,1.5)$ $K/1000 \sim U(0,10)$ $\alpha = \frac{1}{X}\left(\frac{h}{0.2}\right)^{1/0.8}$ $\beta = \frac{ln(h/0.2)}{0.8K}$
A_{ModR}[2]	Modified Ricker	$\alpha SSB_y e^{-\beta (SSB_y)^c}$	$h \sim U(0.2,1.5)$ $K/1000 \sim U(0,10)$ $\alpha = \frac{1}{X}\left(\frac{h}{0.2}\right)^{\frac{1}{1-0.2^c}}$ $\beta = \frac{ln(h/0.2)}{(K)^c[1-0.2^c]}$ $c \sim U(0,1)$
A_{HS}	Hockey stick	$\begin{cases} \alpha & \text{if } SSB_y \geq \beta \\ \alpha \frac{SSB_y}{\beta} & \text{if } SSB_y < \beta \end{cases}$	$ln(\alpha) \sim U(0,7.2)$[3] $\beta/K \sim U(0,1)$ $K = \alpha X$[4]
A_{2HS}	Hockey stick (2 curves)	$\begin{cases} \alpha_1 & \text{if } SSB_y \geq \beta_1 \\ \alpha_1 \frac{SSB_y}{\beta_1} & \text{if } SSB_y < \beta_1 \end{cases}$ if $y < 1999$ $\begin{cases} \alpha_2 & \text{if } SSB_y \geq \beta_2 \\ \alpha_2 \frac{SSB_y}{\beta_2} & \text{if } SSB_y < \beta_2 \end{cases}$ if $y \geq 1999$	$ln(\alpha_t) \sim U(0,7.2)$[4] $\beta_t/K_t \sim U(0,1)$ $K_t = \alpha_t X$[5] $t = 1,2$

[2] Results not shown as c is estimated at 1.
[3] Given the lack of *a priori* information on the scale of a^A, a log-scale was used, with a maximum corresponding to about 10 million tons.
[4] For consistency, K relates throughout to corresponding means.

Table 2. The contributions to the negative log posterior pdf at the joint posterior mode for a range of combinations of juvenile, $\bar{M}_j$, and adult, $\bar{M}_{ad}$, natural mortality for models assuming the Beverton Holt stock recruitment relationship. The ratio of the multiplicative bias in the recruit survey to that in the November survey, $\frac{k_r}{k_N}$, is constrained to be less than 1.

			-lnLikelihood						-lnPriors						Survey bias		
$\bar{M}_j$	$\bar{M}_{ad}$	$-lnPosterior$	$-lnL$	$-lnL^{Nov}$	$-lnL^{Egg}$	$-lnL^{rec}$	$-lnL^{sur}$	$-lnL^{com}$	ε_y	All growth parameters	δ_q	$N_{1983,a}$	η_y^j	k_N	k_N	k_r	$\frac{k_r}{k_N}$
0.6	0.6	-778.65	-837.62	10.14	8.60	39.76	-548.56	-347.56	41.30	4.12	0.52	13.79	0.00	-1.04	0.93	0.93	1.00
0.9	0.6	-780.12	-839.81	10.14	8.57	37.87	-548.51	-347.88	42.14	4.13	0.52	13.76	0.00	-1.16	0.92	0.92	1.00
0.9	0.9	-791.06	-850.64	3.03	6.57	36.09	-547.33	-349.01	42.49	4.30	0.52	13.44	0.00	-1.32	0.89	0.81	0.92
1.2	0.6	-781.05	-841.54	10.17	8.54	36.58	-548.67	-348.18	43.06	4.15	0.52	13.74	0.00	-1.26	0.90	0.90	1.00
1.2	0.9	-790.55	-849.87	3.13	6.48	36.13	-547.31	-348.29	42.29	4.25	0.52	13.43	0.00	-1.33	0.89	0.74	0.83
1.2	**1.2**	**-793.95**	**-851.73**	**-1.33**	**5.15**	**36.46**	**-542.68**	**-349.33**	**40.51**	**4.58**	**0.51**	**13.35**	**0.00**	**-1.31**	0.89	0.61	0.69
1.5	0.6	-781.33	-842.36	10.42	8.55	36.07	-549.23	-348.16	43.71	4.15	0.52	13.70	0.00	-1.34	0.89	0.87	0.99
1.5	0.9	-791.03	-851.06	3.41	6.56	35.82	-547.50	-349.35	42.98	4.35	0.51	13.41	0.00	-1.34	0.88	0.67	0.76
1.5	1.2	-793.51	-851.61	-1.26	5.16	36.34	-542.09	-349.77	40.75	4.69	0.50	13.37	0.00	-1.33	0.89	0.56	0.63
1.5	1.5	-793.58	-850.09	-3.78	3.88	37.04	-537.57	-349.67	38.60	4.95	0.50	13.43	0.00	-1.23	0.91	0.49	0.55
1.8	0.6	-781.34	-842.49	10.67	8.57	35.91	-549.31	-348.33	43.88	4.16	0.51	13.69	0.00	-1.35	0.88	0.79	0.90
1.8	0.9	-790.96	-851.30	3.66	6.55	35.69	-547.71	-349.49	43.29	4.37	0.50	13.41	0.00	-1.35	0.88	0.61	0.69
1.8	1.2	-793.59	-852.22	-0.94	5.11	36.24	-543.16	-349.49	41.27	4.62	0.50	13.39	0.00	-1.33	0.89	0.51	0.57
1.8	1.5	-793.21	-850.06	-3.65	3.87	36.97	-537.52	-349.74	38.83	4.95	0.50	13.50	0.00	-1.25	0.90	0.45	0.50
1.8	1.8	-791.51	-844.62	-4.58	3.34	37.82	-531.50	-349.70	34.68	5.19	0.52	13.51	0.00	-1.16	0.92	0.41	0.45
A_{Mj} with $\sigma_j = 0.2$		-825.56	-850.27	-1.29	5.31	37.00	-541.72	-349.57	31.46	4.65	0.50	13.36	-24.08	-1.29	0.90	0.61	0.68
A_{Mj} with $\sigma_j = 0.1$		-851.35	-851.53	-1.36	5.17	36.48	-542.50	-349.32	38.95	4.58	0.51	13.35	-56.04	-1.30	0.89	0.62	0.69
A_{Mad} with $\sigma_{ad} = 0.2$		-829.19	-866.86	-1.63	5.68	33.17	-555.77	-348.31	42.80	4.83	0.51	13.38	-22.66	-1.36	0.88	0.61	0.69
A_{Mad} with $\sigma_{ad} = 0.1$		-852.70	-856.59	-1.44	5.43	35.05	-546.54	-349.09	41.44	4.65	0.51	13.36	-54.86	-1.33	0.89	0.61	0.69

Table 3. The contributions to the negative log posterior pdf at the joint posterior mode for a range of alternative assumed stock recruitment relationships.

		-lnLikelihood						-lnPriors					
	$-lnPosterio$	$-lnL$	$-lnL^{Nov}$	$-lnL^{Egg}$	$-lnL^{rec}$	$-lnL^{sur}$	$-lnL^{com}$	ε_y	All growth parame ters	δ_q	$N_{1983,a}$	η_y^j	k_N
A_{BH}	-793.95	-851.73	-1.33	5.15	36.46	-542.68	-349.33	40.51	4.58	0.51	13.35	0.00	-1.31
A_{2BH}	-798.61	-850.81	1.13	4.82	36.96	-544.80	-348.92	34.88	4.63	0.51	13.36	0.00	-1.32
A_{3BH}	-800.04	-850.08	0.86	5.12	36.20	-543.33	-348.94	32.79	4.61	0.51	13.36	0.00	-1.35
A_{HS}	-794.05	-851.93	-1.07	5.03	36.91	-543.45	-349.35	40.54	4.60	0.51	13.36	0.00	-1.26
A_{2HS}	-798.61	-850.81	1.13	4.82	36.96	-544.80	-348.92	34.88	4.63	0.51	13.36	0.00	-1.32
A_R	-792.72	-850.65	-1.23	5.23	36.50	-542.47	-348.68	40.71	4.53	0.51	13.35	0.00	-1.31
A_{ModR}	-793.74	-851.05	-1.36	5.11	36.44	-542.49	-348.75	40.09	4.52	0.51	13.35	0.00	-1.31
A_{t0}	-853.74	-891.67	-0.80	5.36	34.61	-576.92	-353.92	44.26	-19.09	0.49	13.49	0.00	-1.30
A_{sel}	-794.98	-853.24	-1.25	5.19	36.42	-543.30	-350.29	40.83	4.64	0.55	13.36	0.00	-1.30

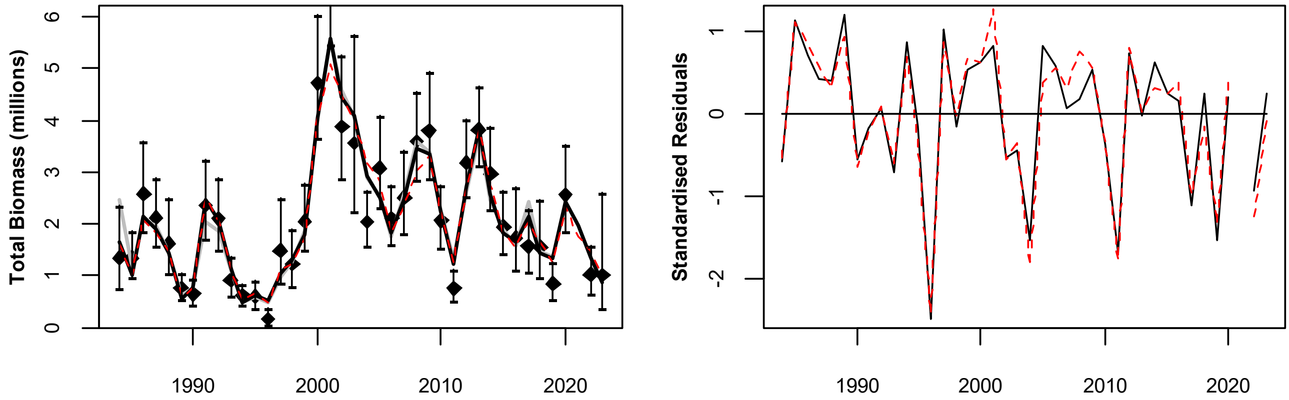


Figure 1. Acoustic November survey and bias corrected model estimated anchovy biomass from 1984 to 2023 for A_{BH} (black) and A_{t0} (red); the 2023 values correspond to that predicted and surveyed in February 2024. The survey indices are shown with 95% confidence intervals reflecting survey inter-transect variance. The standardised residuals are given in the right-hand plot. The A_0 November anchovy biomass estimated by de Moor (2020a) is shown in grey.

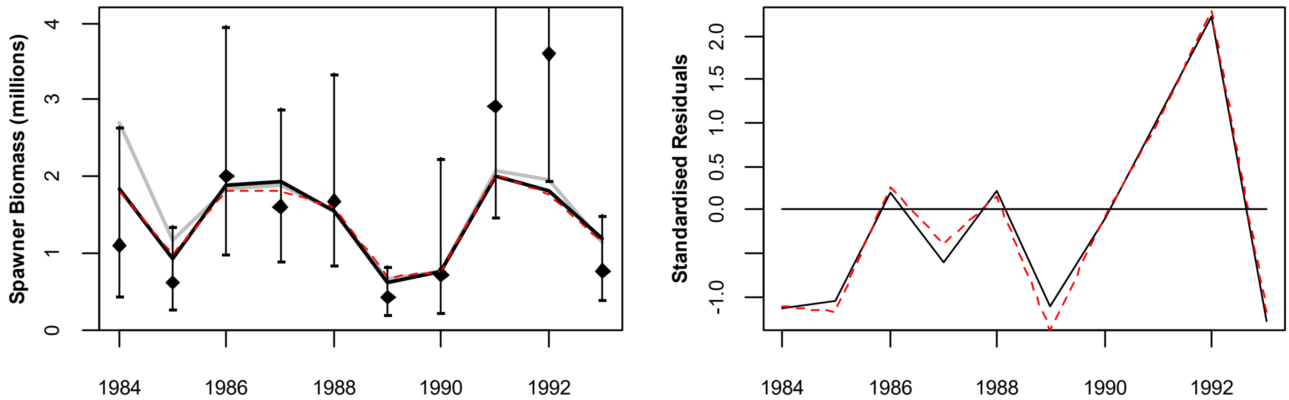


Figure 2. Daily egg production method and bias corrected model estimated anchovy spawner biomass from 1984 to 1993 for A_{BH} (black) and A_{t0} (red). The survey indices are shown with 95% confidence intervals. The standardised residuals are given in the right-hand plot. The A_0 anchovy spawner biomass estimated by de Moor (2020a) is shown in grey.

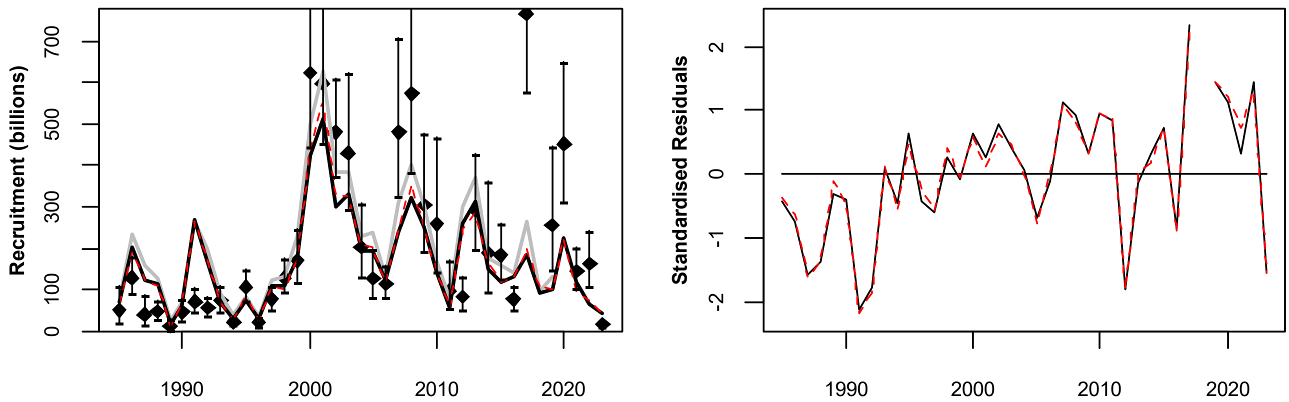


Figure 3. Acoustic survey and bias corrected model estimated anchovy recruitment numbers from 1985 to 2023 for A_{BH} (black) and A_{t0} (red). The survey indices are shown with 95% confidence intervals reflecting survey inter-transect. Note that additional survey variance is estimated included in the model, and not shown in this figure. The standardised residuals are given in the right-hand plot. The A_0 anchovy recruitment estimated by de Moor (2020a) is shown in grey.

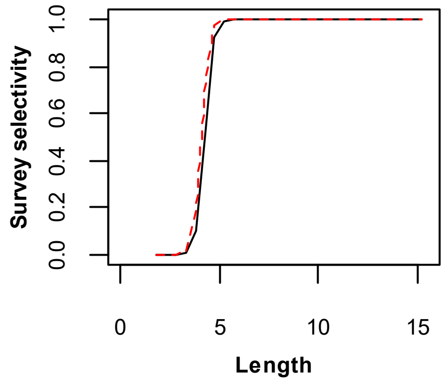


Figure 4. Model estimated trawl survey selectivity at length for A_{BH} (black) and A_{t0} (red).

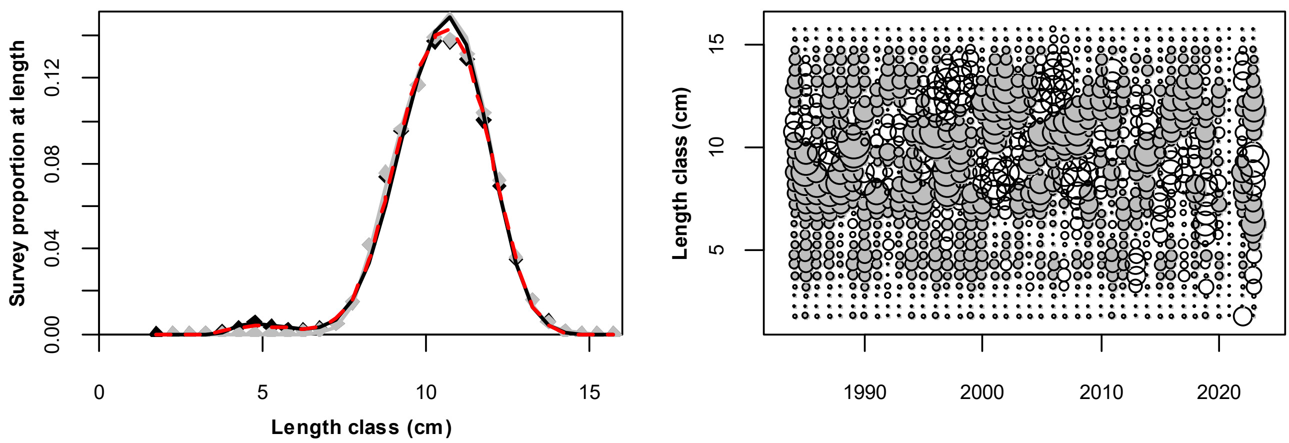


Figure 5. Average (over all years) model predicted and observed proportions-at-length in the November survey trawls for A_{BH} (black) and A_{t0} (red). The standardised residuals for A_{BH} are given in the bubble plot; the size of the bubbles are proportional to the absolute value of the residuals, while the shaded bubbles show negative and the unshaded bubbles show positive residuals.

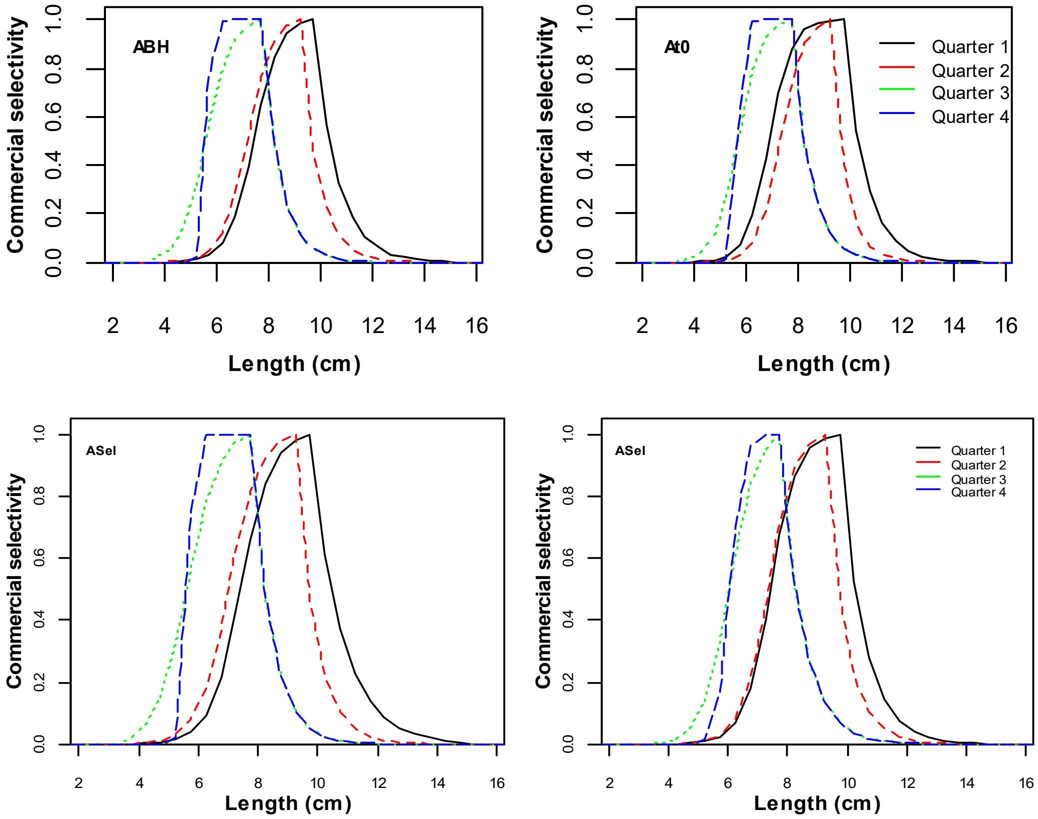


Figure 6. Model estimated quarterly commercial selectivity at length for A_{BH} (top left) and A_{t0} (top right), and A_{sel} (1984-2003 lower left; 2004-2023 lower right).

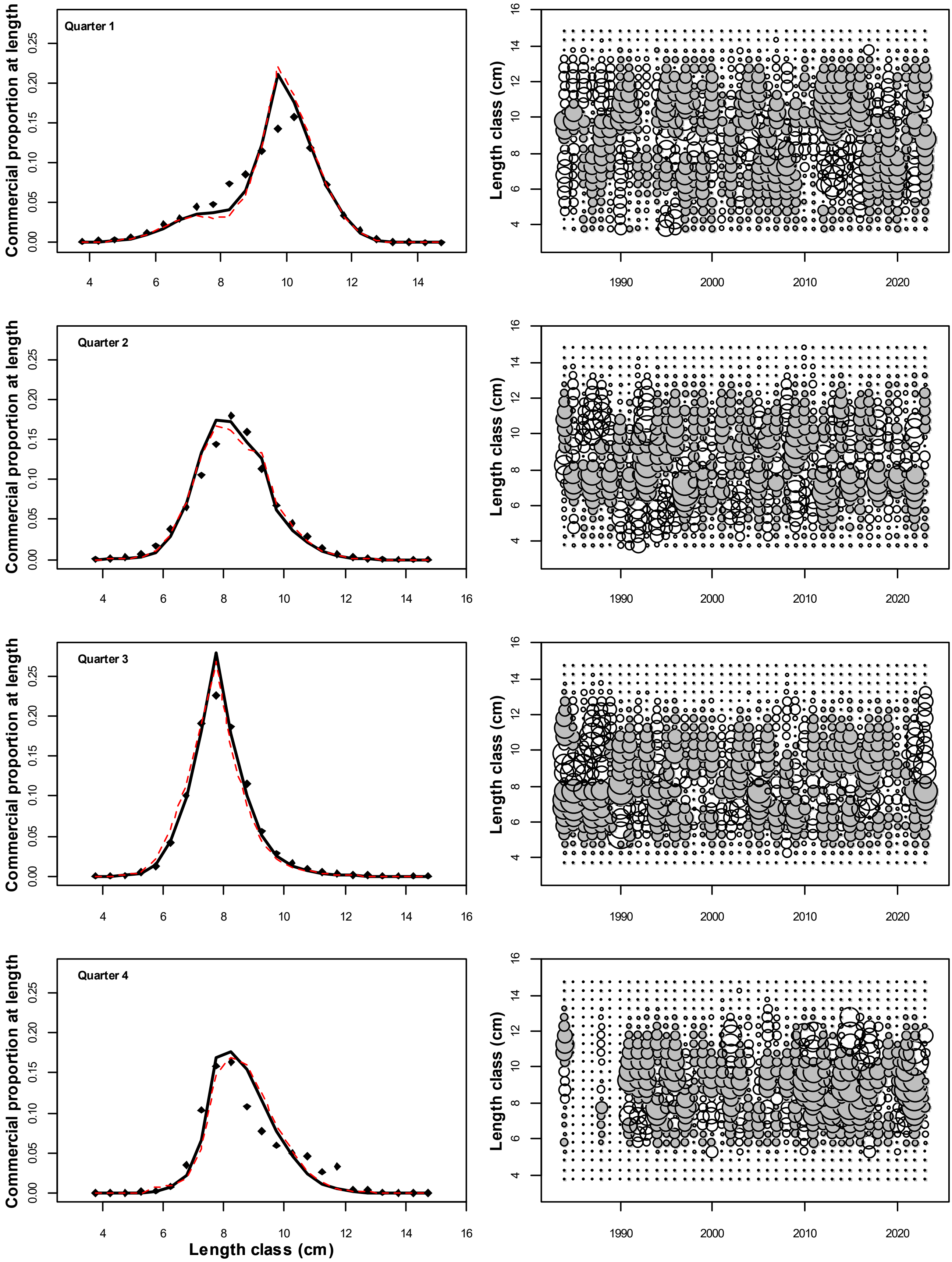


Figure 7. Average (over all years) model predicted and observed proportions-at-length in the quarterly commercial catch for A_{BH} (black) and A_{t0} (red). The standardised residuals for A_{BH} are given in the bubble plot; the size of the bubbles are proportional to the absolute value of the residuals, while the shaded bubbles show negative and the unshaded bubbles show positive residuals.

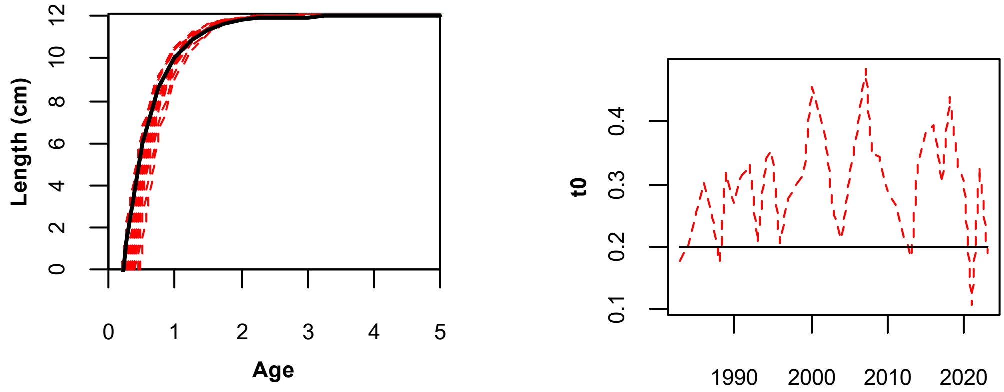


Figure 8. The model estimated von Bertalanffy growth curve(s), where integer ages are taken to correspond to 1 October for A_{BH} (black) and A_{t0} (red). The right hand plot shows the annually estimated $t_{0,y}$ for A_{t0}, with the black line showing $t_0 = 0.2$ estiamated for A_{BH}.

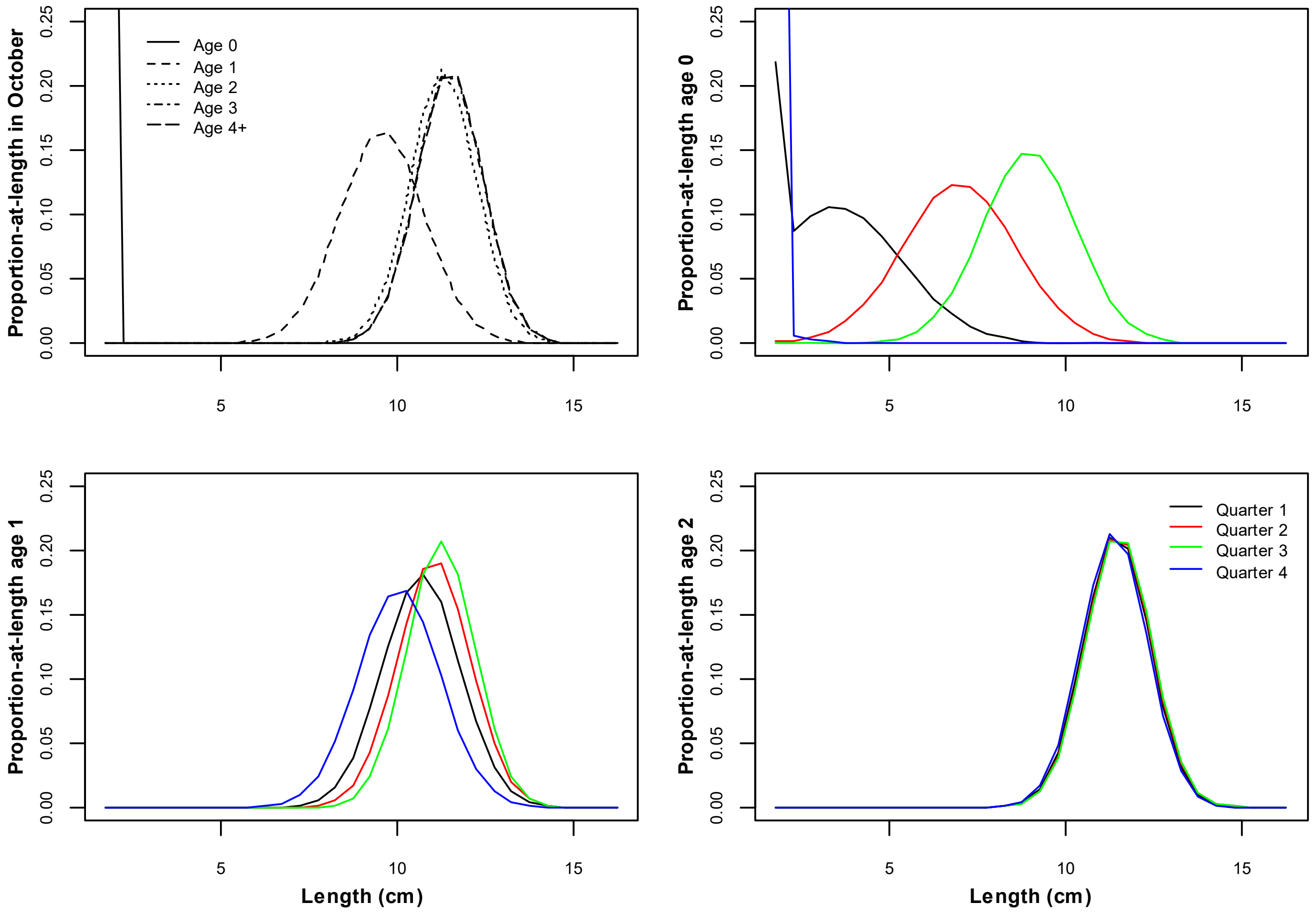


Figure 9. The model estimated distributions of proportions-at-length for each ages 0, 1 and 2, given at the middle of each quarter of the year (corresponding to the times commercial catch is modelled to be taken) and the distributions for all ages at 1 October (corresponding to the time of spawning) for A_{BH}.

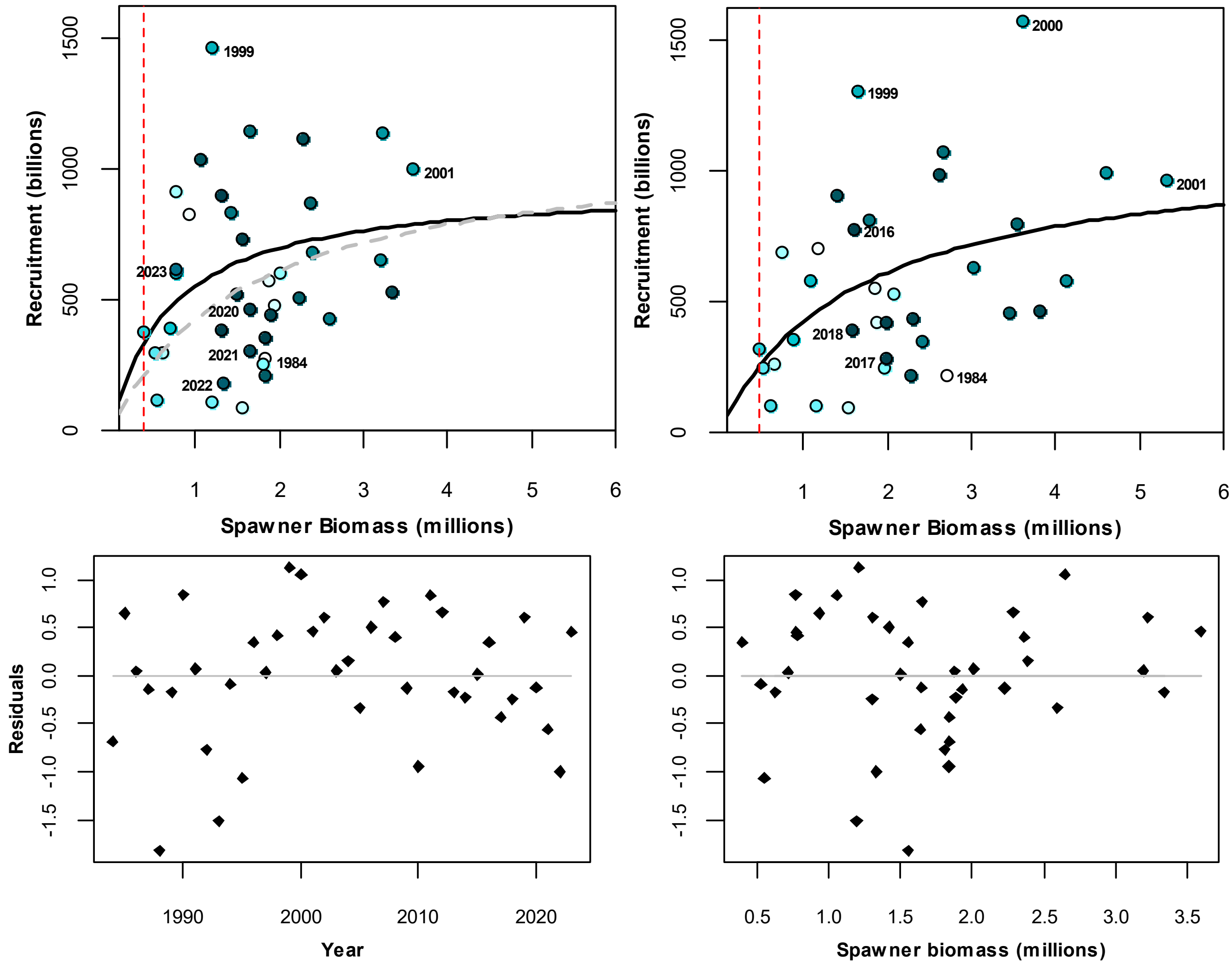


Figure 10. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2023 with the Beverton Holt stock recruitment relationship for A_{BH}. The top right plot shows the relationship estimated by de Moor (2020a). The standardised residuals from the fit are given in the lower plots, against year and against spawner biomass. Note that the final year recruitment may not be well estimated.

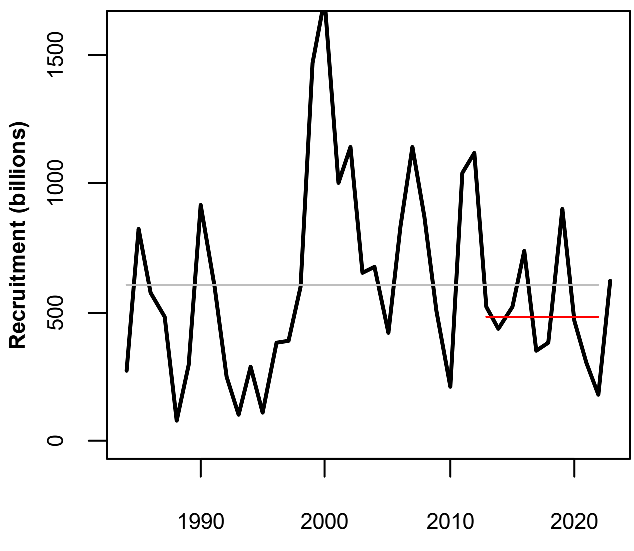


Figure 11. The timeseries of November recruitments for A_{BH}. The grey and red lines indicate the average over the full timeseries (610 billion) and the most recent 10 years (481 billion), respectively, excluding November 2023.

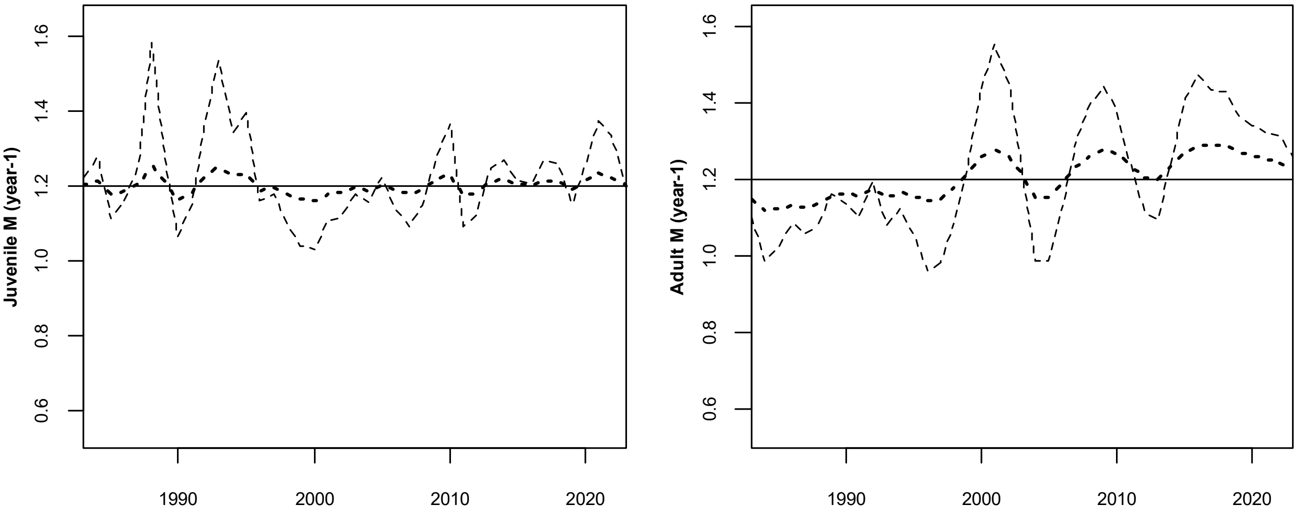


Figure 12. The estimated annual juvenile (left) and adult (right) natural mortality rates for A_{Mj} with $\sigma_j = 0.2$ (dashed) or $\sigma_j = 0.1$ (dotted) and A_{Mad} with $\sigma_{ad} = 0.2$ (dashed) or $\sigma_{ad} = 0.1$ (dotted).

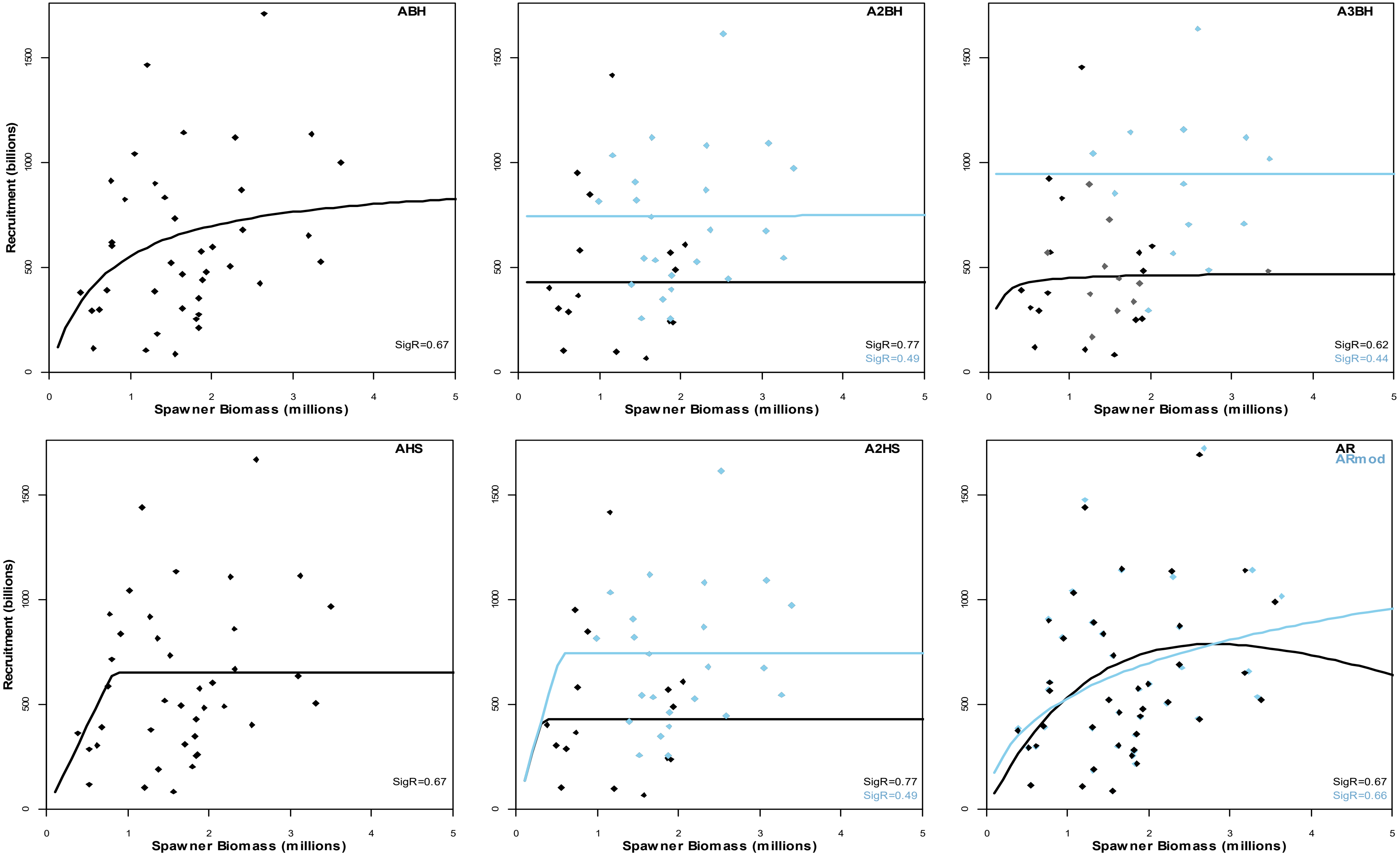


Figure 13. Model predicted anchovy recruitment plotted against spawner biomass from October 1984 to 2023 for the alternative stock recruitment relationships. The blue curves/points denote the years 1999-2012/2023 in cases where more than one relationship is estimated. The estimated standard deviation about the relationship is given in the lower right corner.

Appendix A: Bayesian operating model for the South African anchovy resource

In the below equations a "$\hat{\square}$" is used to represent an estimate of a quantity (e.g. biomass) from a source external to this model (e.g. a survey). Model predicted quantities are represented by terms without any additional super-/sub-scripts other than dependencies on, for example, year, length etc.

Model Assumptions

1) Annual quarters are 1 January – 31 March, 1 April – 30 June, 1 July – 30 September and 1 October – 31 December.
2) Spawning and recruitment are assumed to occur at the start of the 4th quarter (1 October).
3) Anchovy mature according to a length-based ogive with an annually varying L_{50}.
4) A plus group of age 4 is used.
5) Two hydro-acoustic surveys are held each year: the first takes place in November and provides an index of abundance of the total biomass; the second is in May/June (known as the recruit survey) and provides an index of abundance of recruitment.
6) Both surveys provide relative indices of abundance.
7) The November survey is assumed to provide an index of abundance corresponding to 15th November (Figure A.1 of de Moor 2022). The 4th quarter catch is modelled to be taken in a pulse immediately before this estimate of biomass is calculated.
8) The recruit survey estimate of recruit abundance is assumed to correspond with recruitment on the day on which the survey began.
9) The egg survey observations (derived from data collected during the earlier November surveys) provide estimates of spawner biomass in absolute terms.
10) Pulse fishing occurs four times a year, in the middle of each quarter.

Population Dynamics

The assessment is begun at the beginning of the 4th quarter of $y_0 = 1983$, but conditioned to data from $y_1 = 1984$ to $y_n = 2023$. The following subscript notation is used:

- quarters $q = 1$ denoting January to March, $q = 2$ denoting April to June, $q = 3$ denoting July to September and $q = 4$ denoting October to December;
- ages $a = 0$ to a plus group of $a = 4^+$;
- lengths from a minus group of $l = 1.5^-$cm to a plus group of $l = 16^+$cm, with length classes of size 0.5cm;

Numbers-at-age 1+ at 1 October

$$N_{y,a} = \left(\left(\left(\left(N_{y-1,a-1}e^{-M_{a-1,y}/8} - C_{y-1,4,a-1}\right)e^{-M_{a-1,y}/4} - C_{y,1,a-1}\right)e^{-M_{a-1,y}/4} - C_{y,2,a-1}\right)e^{-M_{a-1,y}/4} - C_{y,3,a-1}\right)e^{-M_{a-1,y}/8}$$

$$y_1 \le y \le y_n\,,\ 1 \le a \le 3$$

$$N_{y,4+}=\left(\left(\left(\left(N_{y-1,3}e^{-M_{3,y}/8}-C_{y-1,4,3}\right)e^{-M_{3,y}/4}-C_{y,1,3}\right)e^{-M_{3,y}/4}-C_{y,2,3}\right)e^{-M_{3,y}/4}-C_{y,3,3}\right)e^{-M_{3,y}/8}$$
$$+\left(\left(\left(\left(N_{y-1,4+}e^{-M_{4+,y}/8}-C_{y-1,4,4+}\right)e^{-M_{4+,y}/4}-C_{y,1,4+}\right)e^{-M_{4+,y}/4}-C_{y,2,4+}\right)e^{-M_{4+,y}/4}-C_{y,3,4+}\right)e^{-M_{4+,y}/8}$$
$y_1 \le y \le y_n$ (A.1) [5]

Natural mortality

$M_{0,y}=\bar{M}_j e^{\varepsilon_{j,y}}$ with $\varepsilon^j_{1984}=\eta^j_{1984}$ and $\varepsilon^j_y=\rho\varepsilon^j_{y-1}+\eta^j_y\sqrt{1-\rho^2}$ $\quad y_1 \le y \le y_{n+1}$ (A.2)

$M_{1+,y}=\bar{M}_{ad} e^{\varepsilon_{ad,y}}$ with $\varepsilon^{ad}_{1984}=\eta^{ad}_{1984}$ and $\varepsilon^{ad}_y=\rho\varepsilon^{ad}_{y-1}+\eta^{ad}_y\sqrt{1-\rho^2}$ $\quad y_1 \le y \le y_{n+1}$ (A.3)

Numbers-at-length

The model predicted numbers-at-length of ages 1+ only at 1st October are given by:

$N^{1+}_{y,l}=\sum_{a=1}^{4+}A^{Oct}_{a,l}N_{y,a}$ $\quad y_1 \le y \le y_n$, $1.5^-cm \le l \le 16^+cm$ (A.4)

Numbers-at-length at 15th November, after catch

$$N_{y,l}=\begin{cases}\sum_{a=0}^{4+}A_{4,a,l}\left(N_{y,a}e^{-M_{a,y+1}/8}-C_{y,4,a}\right) & y_1 \le y \le y_{n-1}\\ \sum_{a=0}^{4+}A_{1,a,l}\left[\left(N_{y,a}e^{-M_{a,y+1}/8}-C_{y,4,a}\right)e^{-7M_{a,y+1}/24}\right] & y=y_n\end{cases}$$
$1.5^-cm \le l \le 16^+cm$ (A.5) [6]

Growth

The proportion of anchovy of age a that fall in the length group l is calculated under the assumption that length-at-age is normally distributed about a Richards or von Bertalanffy (if $D=0$) growth curve. This is calculated at times during the year modelled to correspond to spawning (1 October) and commercial catches (midpoint of each quarter):

$A^{Oct}_{a,l}\sim N\left(L_\infty\left(1+(D-1)e^{-\kappa(a-t0)}\right)^{1/(1-D)},\vartheta^2_a\right)$ $\quad 0 \le a \le 4^+$, $1.5^-cm \le l \le 16^+cm$ (A.6)[7]

$$A_{q,a,l}\sim\begin{cases}N\left(L_\infty\left(1+(D-1)e^{-\kappa(a+(2q+1)/8-t0)}\right)^{1/(1-D)},\left(1-\frac{2q+1}{8}\right)\vartheta^2_a+\frac{2q+1}{8}\vartheta^2_{a+1}\right) & 1 \le q \le 3\\ N\left(L_\infty\left(1+(D-1)e^{-\kappa(a+1/8-t0)}\right)^{1/(1-D)},\left(1-\frac{1}{8}\right)\vartheta^2_a+\frac{1}{8}\vartheta^2_{a+1}\right) & q=4\end{cases}$$
$0 \le a \le 4^+$, $1.5^-cm \le l \le 16^+cm$ (A.7)[8]

Spawning biomass at 1 October

$SSB_y=\sum_{l=1.5^-}^{16^+}f_{y,l}N^{1+}_{y,l}w_l$ $\quad y_1 \le y \le y_n$ (A.8)

where

$f_{y,l}=1/\left(1+e^{-(l-L_{50,y})/\delta^{mat}_y}\right)$ $\quad y_1 \le y \le y_n$ (A.9)[9]

[5] $C_{1983,4,a}=0$ as $C^{RLF}_{1983,m,l}$ is unknown.
[6] This adjustment for $y=y_n$ is to 1st March 2024 for the delayed November 2023 survey which took place from mid-February to mid-March 2024; minimal anchovy catch was landed between January and March 2024 and is thus ignored.
[7] The proportion is calculated as the area under the curve between the lower limit and upper limit of length class l. The lower and upper tails are included in the proportions calculated for the minus and plus groups, respectively.
[8] The proportion is calculated as the area under the curve between the lower limit and upper limit of length class l. The lower and upper tails are included in the proportions calculated for the minus and plus groups, respectively.
[9] Where length is used in an equation, the mid-point of the length class is used.

Recruitment at 1 October

$$N_{y,0} = f\left(SSB_y\right)e^{\varepsilon_y - 0.5\sigma_r^2} \qquad y_1 \le y \le y_n \text{ (A.10)}$$

Biomass associated with the November survey

$$B_y = k_N \sum_{l=1.5^-}^{16^+} N_{y,l} w_l \qquad y_1 \le y \le y_n \text{ (A.11)}$$

Proportion-at-length associated with the November survey

$$p_{y,l} = \frac{N_{y,l} S_l^{sur}}{\sum_{l=2}^{15.5} N_{y,l} S_l^{sur}} \qquad y_1 \le y \le y_n, \text{ (A.12)}$$

where

$$S_l^{sur} = \begin{cases} 0 & l = 1.5^- \\ \frac{1}{1+exp(-(l-l^{sur})/\delta^{sur})} & 2cm \le l \le 15.5cm \\ 0 & l = 16^+ \end{cases} \qquad \text{(A.13)}^{13}\ ^{10}$$

Commercial selectivity

Commercial selectivity-at-length is assumed to follow the logistic shape, with a dome at high lengths. Commercial selectivity is assumed to vary by quarter, but remain unchanged over time. Selectivity-at-lengths less than the smallest observed length class (3.5cm) and greater than the largest observed length class (14.5cm) are taken to be zero:

$$S_{y,q,l} = \begin{cases} 0 & 1.5^- cm \le l \le 3cm \\ 1/\left(1+e^{\psi_q(l - l50_q)}\right) & 3.5cm \le l \le S_q^{break} \\ S_{y,q,l-1}e^{\delta_q} & S_q^{break} \le l \le 14.5cm \\ 0 & 15cm \le l \le 16^+ cm \end{cases} \qquad y_1 \le y \le y_n, 1 \le q \le 4 \text{ (A.14)}^{13}\ ^{11}$$

Commercial selectivity-at-age is given by:

$$S_{y,q,a} = \sum_{l=1.5^-}^{16^+} A_{q,a,l} S_{y,q,l} \qquad y_1 \le y \le y_n, 1 \le q \le 4, 0 \le a \le 4^+ \text{ (A.15)}$$

Commercial catch

$$C_{y,4,a} = N_{y,a} e^{-M_{a,y+1}/8} S_{y,4,a} F_{y,4}$$

$$C_{y,1,a} = \left(N_{y-1,a} e^{-M_{a,y}/8} - C_{y-1,4,a}\right) e^{-M_{a,y}/4} S_{y,1,a} F_{y,1}$$

$$C_{y,2,a} = \left(\left(N_{y-1,a} e^{-M_{a,y}/8} - C_{y-1,4,a}\right) e^{-M_{a,y}/4} - C_{y,1,a}\right) e^{-M_{a,y}/4} S_{y,2,a} F_{y,2}$$

$$C_{y,3,a} = \left(\left(\left(N_{y-1,a} e^{-M_{a,y}/8} - C_{y-1,4,a}\right) e^{-M_{a,y}/4} - C_{y,1,a}\right) e^{-M_{a,y}/4} - C_{y,2,a}\right) e^{-M_{a,y}/4} S_{y,3,a} F_{y,3}$$

$$y_1 \le y \le y_n, 0 \le a \le 4^+ \text{ (A.16)}$$

[10] A zero survey trawl selectivity is assumed for lengths 1.5^-cm and 16^+cm, consistent with the range of length classes in the observed trawl survey proportions-at-length.

[11] These selectivities-at-length are renormalized so that the maximum is 1.

The fished proportion of the available biomass from the anchovy fishery is estimated by:

$$F_{y,4} = \frac{\sum_{m=10}^{12}\sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4^+} N_{y,a}e^{-M_{a,y+1}/8}S_{y,4,a}}$$

$$F_{y,1} = \frac{\sum_{m=1}^{3}\sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4^+}\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4}S_{y,1,a}}$$

$$F_{y,2} = \frac{\sum_{m=4}^{6}\sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4^+}\left(\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4} - C_{y,1,a}\right)e^{-M_{a,y}/4}S_{y,2,a}}$$

$$F_{y,3} = \frac{\sum_{m=7}^{9}\sum_{l=3.5}^{14.5} C_{y,m,l}^{RLF}}{\sum_{a=0}^{4^+}\left(\left(\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4} - C_{y,1,a}\right)e^{-M_{a,y}/4} - C_{y,2,a}\right)e^{-M_{a,y}/4}S_{y,3,a}} \qquad y_1 \le y \le y_n \text{ (A.17)}$$ [12]

A penalty is imposed within the model to ensure that $S_{y,q,l}F_{y,q} < 0.95$ for all l.

Number of recruits at the time of the recruit survey

$$N_{y,r} = k_r\left(\left(\left(N_{y-1,0}e^{-M_{0,y}/8} - C_{y-1,4,0}\right)e^{-M_{0,y}/4} - C_{y,1,0}\right)e^{-(1/8+0.5(1+t_y)/12)M_{0,y}} - C_{y,0bs}\right)e^{-(0.5(1+t_y)/12)M_{0,y}}$$

$$y_2 \le y \le y_n \text{ (A.18)}$$

The recruit catch from 1 April to the day before the survey is calculated as follows

$$C_{y,0bs} = \left(\left(N_{y-1,0}e^{-M_{0,y}/8} - C_{y-1,4,0}\right)e^{-M_{0,y}/4} - C_{y,1,0}\right)e^{-(1/8+0.5(1+t_y)/12)M_{0,y}}S_{y,2,0}F_{y,bs} \qquad y_2 \le y \le y_n \text{ (A.19)}$$

where

$$F_{y,bs} = \frac{\sum_{l=3.5}^{14.5} C_{y,bs,l}^{RLF}}{\sum_{a=0}^{4^+}\left(\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4} - C_{y,1,a}\right)e^{-(1/8+0.5(1+t_y)/12)M_{a,y}}S_{y,2,a}} \qquad y_2 \le y \le y_n \text{ (A.20)}$$

A penalty is imposed within the model to ensure that $S_{y,2,l}F_{y,bs} < 0.95$ for all l.

Commercial catch-at-length from the anchovy fishery

$$C_{y,4,l} = \sum_{a=0}^{4+} N_{y,a}e^{-M_{a,y+1}/8}A_{4,a,l}S_{y,4,l}F_{y,4}$$

$$C_{y,1,l} = \sum_{a=0}^{4+}\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4}A_{1,a,l}S_{y,1,l}F_{y,1}$$

$$C_{y,2,l} = \sum_{a=0}^{4+}\left(\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4} - C_{y,1,a}\right)e^{-M_{a,y}/4}A_{2,a,l}S_{y,2,l}F_{y,2}$$

$$C_{y,3,l} = \sum_{a=0}^{4+}\left(\left(\left(N_{y-1,a}e^{-M_{a,y}/8} - C_{y-1,4,a}\right)e^{-M_{a,y}/4} - C_{y,1,a}\right)e^{-M_{a,y}/4} - C_{y,2,a}\right)e^{-M_{a,y}/4}A_{3,a,l}S_{y,3,l}F_{y,3}$$

$$y_1 \le y \le y_n,\ 1.5^-cm \le l \le 16^+cm \text{ (A.21)}$$

Proportion-at-length associated with the commercial catch

$$p_{y,q,l}^{coml} = \frac{C_{y,q,l}}{\sum_{l=3.5}^{14.5} C_{y,q,l}} \qquad y_1 \le y \le y_n,\ 1 \le q \le 4,\ 3.5cm \le l \le 14.5cm \text{ (A.22)}$$ [13]

Fitting the Model to Observed Data (Likelihood)

The survey observations of abundance are assumed to be log-normally distributed. The standard errors of the log-distributions for the survey observations of adult biomass and recruitment numbers are approximated by the CVs of the untransformed

[12] The range of length classes used in these summation matches the range of length classes in the observations which is a smaller range than the maximum range modelled of 1.5^-cm to 16^+cm.

[13] Note the model predicted commercial catch of lengths <3.5cm and >14.5cm are zero (equation A.14), consistent with the range of length classes observed in the commercial catch.

distributions and a further additional variance parameter. A "sqrt(p)" formulation, rather than the "adjusted lognormal" ("Punt-Kennedy", Punt and Kennedy 1997) error distribution formulation, is assumed for the estimated proportions-at-length particularly as it can deal with occasional zero observations more easily. This "sqrt(p)" formulation mimics a multinomial form for the error distribution by forcing near-equivalent variance-mean relationship for the error distributions. The negative log-likelihood function is given by:

$$-lnL = -lnL^{Nov} - lnL^{Egg} - lnL^{rec} - lnL^{surpropl} - lnL^{compropl} \tag{A.23}$$

where

$$-lnL^{Nov} = \frac{1}{2}\sum_{y=y_1}^{y_n}\left\{\frac{\left(ln\hat{B}_y - ln(B_y)\right)^2}{\sigma_{y,N}^2+\phi_{ac}^2+\lambda_N^2} + ln\left(2\pi\left(\sigma_{y,N}^2 + \phi_{ac}^2 + \lambda_N^2\right)\right)\right\} \tag{A.24}$$ [14]

$$-lnL^{Egg} = \frac{1}{2}\sum_{y=y_1}^{1993}\left\{\frac{\left(ln\hat{B}_{y,egg} - ln\left(k_g SSB_y\right)\right)^2}{\sigma_{y,egg}^2} + ln\left(2\pi\sigma_{y,egg}^2\right)\right\} \tag{A.25}$$

$$-lnL^{rec} = \frac{1}{2}\sum_{y=y_1+1}^{y_n}\left\{\frac{\left(ln\hat{N}_{y,r} - ln(N_{y,r})\right)^2}{\sigma_{y,r}^2+\phi_{ac}^2+\lambda_r^2} + ln\left(2\pi\left(\sigma_{y,r}^2 + \phi_{ac}^2 + \lambda_r^2\right)\right)\right\} \tag{A.26}$$ [15]

$$-lnL^{surpropl} = w_{propl}^{sur}\sum_{y=y_1}^{y_n}\sum_{l=2}^{15.5}\left\{\frac{\left(\sqrt{\hat{p}_{y,l}} - \sqrt{p_{y,l}}\right)^2}{2\sigma_{sur}^2} + ln(\sigma_{sur})\right\}^{16} \tag{A.27}$$

$$-lnL^{compropl} = w_{propl}^{com}\sum_{y=y_1}^{y_n}\sum_{q=1}^{4}\sum_{l=3.5}^{14.5}\left\{\frac{\left(\sqrt{\hat{p}_{y,q,l}^{coml}} - \sqrt{p_{y,q,l}^{coml}}\right)^2}{2\sigma_{com}^2} + ln(\sigma_{com})\right\} \tag{A.28}$$

[14] Excluding 2021 when no survey was completed.
[15] Excluding 2018 when no survey was completed.
[16] Although strictly there may be bias in the proportions of length-at-age data, no bias is assumed in this assessment. The effect of such a bias is assumed to be small.

Table A.1. Assessment model parameters and variables.

	Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
Annual numbers and biomass	$N_{y,a}$	Model predicted numbers-at-age a at 1 October in year y	Billions		A.1 & A.10	
	$N_{1983,a}$	Initial numbers-at-age a	Billions	$N_{1983,0} \sim N(53, 100^2)$, $N_{1983,1} \sim N(482, 200^2)$, $N_{1983,2} \sim N(1, 2^2)$, $N_{1983,3} = N_{1983,2} e^{-M^A_{2,1983}}$, $N_{1983,4} = N_{1983,3} \times \frac{e^{-M^A_{3,1983}}}{1-e^{-M^A_{3,1983}}}$		Appendix B $M^A_{ad,1983} = M^A_{ad,1984}$
	$N_{y,l}$	Model predicted numbers-at-length l at 15 November in year y	Billions		A.5	
	$N^{1+}_{y,l}$	Model predicted numbers-at-length length l at 1 October in year y of anchovy ages 1+ only	Billions		A.4	
	B_y	Model predicted total biomass at 15 November in year y	Thousand tons		A.11	
	w_l	Mean mass of anchovy of length l[17] (in cm) during November	Grams	$w_l = 0.007366 \times l^{3.11124}$		de Moor (2023)
	SSB_y	Model predicted spawning biomass at 1 October in year y	Thousand tons		A.8	
	$f_{y,l}$	Proportion of anchovy of length l (in cm) that are mature in year y			A.9	
	$L_{50,y}$	Length at 50% maturity in year y	Cm	Table A.3		Yemane *et al*. 2024
	δ^{mat}_y	Rate of increase in maturity at length in year y	-	Table A.3		Yemane *et al*. 2024
Catch	$C_{y,q,a}$	Model predicted number of anchovy of age a caught during quarter q in year y	Billions		A.16	
	$F_{y,q}$	Fished proportion in quarter q of year y for a fully selected length class l	-		A.17	
	$C_{y,0bs}$	Number of recruits caught between 1 April and the day before the start of the recruit survey in year y	Billions		A.19	
	$F_{y,bc}$	Fished proportion between 1 April and the day before the start of the recruit survey in year y	-		A.20	

[17] Where length is required in an equation, the mid-point of the length class is used.

Table A.1 (continued).

	Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
`Natural Mortality	$M_{a,y}$	Rate of natural mortality of age a in year y	Year^{-1}		A.2 & A.3	
	$\bar{M}_j$	Median rate of natural mortality for age-0 anchovy	Year^{-1}	1.2		See alternatives tested in this document
	$\bar{M}_{ad}$	Median rate of natural mortality for 1+ anchovy	Year^{-1}	1.2		
	ε_y^j	Annual residuals about natural mortality rate for age-0 anchovy	-		A.2	
	ε_y^{ad}	Annual residuals about natural mortality rate for 1+ anchovy	-		A.3	
	η_y^j	Normally distributed error in calculating ε_y^j	-	$\sim N(0, \sigma_j^2)$		
	η_y^{ad}	Normally distributed error in calculating ε_y^{ad}	-	$\sim N(0, \sigma_{ad}^2)$		
	σ_j	Standard deviation in the annual residuals η_y^j	-	0.1 or 0.2		See alternatives tested in this document
	σ_{ad}	Standard deviation in the annual residuals η_y^{ad}	-	0.1 or 0.2		
	ρ	Annual autocorrelation coefficient	-	$\sim U(0,1)$		
Selectivity	S_l^{sur}	November survey trawl selectivity-at-length l	-		A.13	
	l^{sur}	Length class number at which the survey selectivity-at-length is 50%	Cm	$U(2.5,11.5)$		
	δ^{sur}	Steepness of the survey selectivity-at-length relationship		$U(0.005,5)$		
	$S_{y,q,l}$	Commercial selectivity-at-length l during quarter q of year y	-		A.14	
	$S_{y,q,a}$	Commercial selectivity-at-age a during quarter q of year y	-		A.15	
	ψ_q	Steepness of ascending limb of logistic part of commercial selectivity curve during quarter q	-	$\sim U(-10,0)$, $\psi_1 = \psi_2 = \psi_3$		Uninformative
	$l50_q$	Length at which ascending limb of logistic part of commercial selectivity is 50% during quarter q	Cm	$\sim U(3,10)$, $l50_3 = l50_4$		Uninformative
	δ_q	Rate of exponential decrease in commercial selectivity at large lengths during quarter q	-	$\delta_1 \sim N(-0.49, 0.5^2)$, $\delta_2 \sim N(-0.75, 1.0^2)$, $\delta_3 = \delta_4 \sim N(-0.66, 0.2^2)$		See Appendix B
	S_q^{break}	Length at which commercial selectivity starts to decrease during quarter q	Cm	$S_1^{break} = 9.5$, $S_2^{break} = 9$, $S_3^{break} = S_4^{break} = 7.5$		

Table A.1 (Continued).

	Parameter/ Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
Recruitment	h	Steepness associated with the stock-recruitment curve[18]	-	Table 1		
	K	Carrying capacity	Thousand tons	Table 1		
	α	Stock-recruitment curve parameter, related to h and K, for Beverton Holt and Ricker curves	-		$=\frac{4hK}{(5h-1)\left(\sum_{a=1}^{3}\bar{f}_a w_a e^{-\overline{M}_j-(a-1)\overline{M}_{ad}}+\bar{f}_{4+}w_{4+}e^{-\overline{M}_j-3\overline{M}_{ad}}/\left(1-e^{-\overline{M}_{ad}}\right)\right)}$	
	β	Stock-recruitment curve parameter, related to h and K, for Beverton Holt and Ricker curves	Thousand tons	$=\frac{(1-h)K}{(5h-1)}$		
	ε_y	Annual lognormal deviation of recruitment	-	$\sim N(0,\sigma_r^2)$, $y_1 \leq y \leq 1999$ $\sim N\left(0,\sigma_{r,2000+}^2\right)$,$2000 \leq y \leq y_n$		
	σ_r^2	Variance in the residuals (lognormal deviation) about the stock recruitment curve	-	$\sim U(0.16,10)$		Lower bound chosen to restrict the influence of the stock recruitment curve on the assessment results
	$\bar{f}_a$	Average proportion of anchovy mature at age a			$=\frac{1}{y_n-y_1+1}\sum_{y=y_1}^{y_n}\sum_{l=1.5^-}^{16^+} f_{y,l}A_{a,l}^{Oct}$	
	w_a	Mean mass at age a at spawning		$=\sum_{l=1.5^-}^{16^+} w_l A_{a,l}^{Oct}$		
	$N_{y,r}$	Model predicted number of recruits at the time of the recruit survey in year y	Billions		A.18	

[18] The proportion of the median virgin recruitment that is realised at a spawning biomass level of 20% of average pre-exploitation (virgin) spawning biomass, K.

Table A.1 (Continued).

	Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
Multiplicative bias	k_N	Multiplicative bias associated with the November acoustic survey	-	$ln(k_N) \sim N(-0.158, 0.112^2)$		de Moor *et al*. (2020a)
	k_g	Multiplicative bias associated with the November egg survey	-	1.0		From de Moor *et al*. (2020b)
	k_r	Multiplicative bias associated with the recruit survey	-	$k_r/k_N \sim U(0,1)$		Recruit survey assumed to cover less of the recruits than the November survey covers of the total biomass
Proportions-at-length and growth	$p_{y,l}$	Model predicted proportion-at-length l associated with the November survey in year y	-		A.12	
	$A_{a,l}^{Oct}$	Proportion of anchovy-at-age a that fall in the length group l at 1 October	-		A.6	
	$p_{y,q,l}^{com}$	Model predicted proportion-at-length l in the commercial catch during quarter q of year y	-		A.22	
	$A_{q,a,l}$	Proportion of anchovy-at-age a that fall in the length group l in quarter q	-		A.7	
	L_∞	Maximum length (in expectation) of anchovy	Cm	$\sim N(12, 1^2)$		Appendix B
	κ	Annual somatic growth rate of anchovy	Year^{-1}	$\frac{\kappa \times L_\infty}{10} \sim N(2.1, 1^2)$		Appendix B
	t_0	Age at which the length (in expectation) is zero for von Bertalanffy growth curve	Year	$\sim N(0.1, 1^2)$		Appendix B
	ϑ_a	Standard deviation of the distribution about the mean length for age a	-	$\vartheta_0 \sim N(2, 1.5^2)$, $\vartheta_1 \sim N(1.3, 1.5^2)$, $\vartheta_{2+} \sim N(1.0, 0.1^2)$		Appendix B

Table A.1 (Continued).

	Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation	Notes
Likelihood	$-lnL^{Nov}$	Contribution to the negative log likelihood from the model fit to the November total survey biomass data	-		A.24	
	$-lnL^{Egg}$	Contribution to the negative log likelihood from the model fit to the egg survey spawner biomass data			A.25	
	$-lnL^{rec}$	Contribution to the negative log likelihood from the model fit to the recruit survey data	-		A.26	
	$-lnL^{surpropL}$	Contribution to the negative log likelihood from the model fit to the November survey proportion-at-length data	-		A.27	
	$-lnL^{compropL}$	Contribution to the negative log likelihood from the model fit to the quarterly commercial proportion-at-length data	-		A.28	
	λ_N^2	Additional variance, over and above $\sigma_{y,N}^2$, associated with the November survey	-	0		See robustness tests
	λ_r^2	Additional variance, over and above $\sigma_{y,r}^2$, associated with the recruit survey		$\sim U(0,10)$		Uninformative
	ϕ_{ac}^2	Additional hydro-acoustic survey variance	-	0.197^2		de Moor *et al.* (2020a)
	w_{prop}^{sur}	Weighting applied to the survey proportion-at-length data	-	0.2		To allow for autocorrelation[19]
	σ_{sur}	Standard deviation associated with the survey proportion-at-length data	-	$\sqrt{\frac{\sum_{y=y_1}^{y_n}\sum_{l=7}^{13}\left(\sqrt{\hat{p}_{y,l}}-\sqrt{p_{y,l}}\right)^2}{\sum_{y=y_1}^{y_n}\sum_{l=7}^{13}1}}$		Closed form solution[20]
	w_{prop}^{com}	Weighting applied to the commercial proportion-at-length data	-	0.05		To allow for autocorrelation[21]
	σ_{com}	Standard deviation associated with the commercial proportion-at-length data	-	$\sqrt{\frac{\sum_{y=y_1}^{y_n}\sum_{q=1}^{4}\sum_{l=5}^{12}\left(\sqrt{\hat{p}_{y,q,l}^{coml}}-\sqrt{p_{y,q,l}^{coml}}\right)^2}{\sum_{y=y_1}^{y_n}\sum_{q=1}^{4}\sum_{l=5}^{12}1}}$		Closed form solution[22]

[19] Based upon data being available ~5 times more frequently than annual age data which contain maximum information content on this
[20] A shorter range of lengths ($7cm \le l \le 13cm$) is used given the near absence of data outside this range, resulting in small/zero residuals, which would negatively bias this estimate.
[21] Based upon data being available ~4x5 times more frequently than annual age data which contain maximum information content on this
[22] A shorter range of lengths ($5cm \le l \le 12cm$) is used given the near absence of data outside this range, resulting in small/zero residuals, which would negatively bias this estimate.

Table A.2. Assessment model data, detailed in de Moor *et al*. (2024).

Quantity	Description	Units / Scale	Shown in Figure
$C_{y,m,l}^{RLF}$	Observed number of anchovy in length class l caught during month m of year y[23]	Billions	
$C_{y,bs}^{RLF}$	Observed number of anchovy in length class l caught from 1 April to the day before the start of the recruit survey in year y	Billions	
t_y	Time lapsed between 1 May and the start of the recruit survey in year y	Months	
$\hat{B}_y$	Acoustic survey estimate of total biomass from the November survey in year y	Thousand tons	Figure 1
$\sigma_{y,N}$	Survey sampling CV associated with $\hat{B}_y$ that reflects survey inter-transect variance	-	Figure 1
$\hat{B}_{y,egg}$	Egg survey estimate of spawner biomass from the November survey in year y	Thousand tons	Figure 2
$\sigma_{y,egg}$	Survey sampling CV associated with $\hat{B}_{y,egg}$ estimated from inter-transect variance		Figure 2
$\hat{N}_{y,r}$	Acoustic survey estimate of recruitment from the recruit survey in year y	Billions	Figure 3
$\sigma_{y,r}$	Survey sampling CV associated with $\hat{N}_{y,r}$ that reflects survey inter-transect variance	-	Figure 3
$\hat{p}_{y,l}$	Observed proportion (by number) of anchovy in length group l in the November survey of year y	-	
$\hat{p}_{y,q,l}^{com}$	Observed proportion (by number) of anchovy commercial catch in length group l during quarter q of year y		

Table A.3. Annual length (in cm) at which maturity is 50% and associated steepness (Yemane *et al*. 2024).

Year	$L_{50,y}$	δ_y^{mat}	Year	$L_{50,y}$	δ_y^{mat}	Year	$L_{50,y}$	δ_y^{mat}
1984[24]	8.891	1.250	1998	10.367	2.668	2012	8.393	1.550
1985	8.891	1.250	1999	9.557	2.678	2013	8.848	1.298
1986	8.495	1.304	2000	9.809	1.551	2014	10.034	1.962
1987	8.169	1.116	2001	10.179	1.833	2015	9.449	1.575
1988	7.987	1.366	2002	10.058	1.592	2016	9.012	0.848
1989	7.639	1.397	2003	9.673	1.414	2017	8.910	1.300
1990	7.793	0.868	2004	8.179	3.605	2018	8.890	1.734
1991	8.461	1.664	2005	9.069	0.359	2019	8.724	0.768
1992	8.887	1.695	2006	9.568	1.979	2020	9.380	2.830
1993	7.691	1.740	2007	9.922	1.730	2021[25]	9.085	1.985
1994	7.528	1.839	2008	10.045	1.340	2022	8.789	1.141
1995	8.547	1.763	2009	10.101	2.327	2023[26]	8.939	1.555
1996	9.406	2.034	2010	9.467	1.873			
1997	9.648	1.732	2011	8.292	2.906			

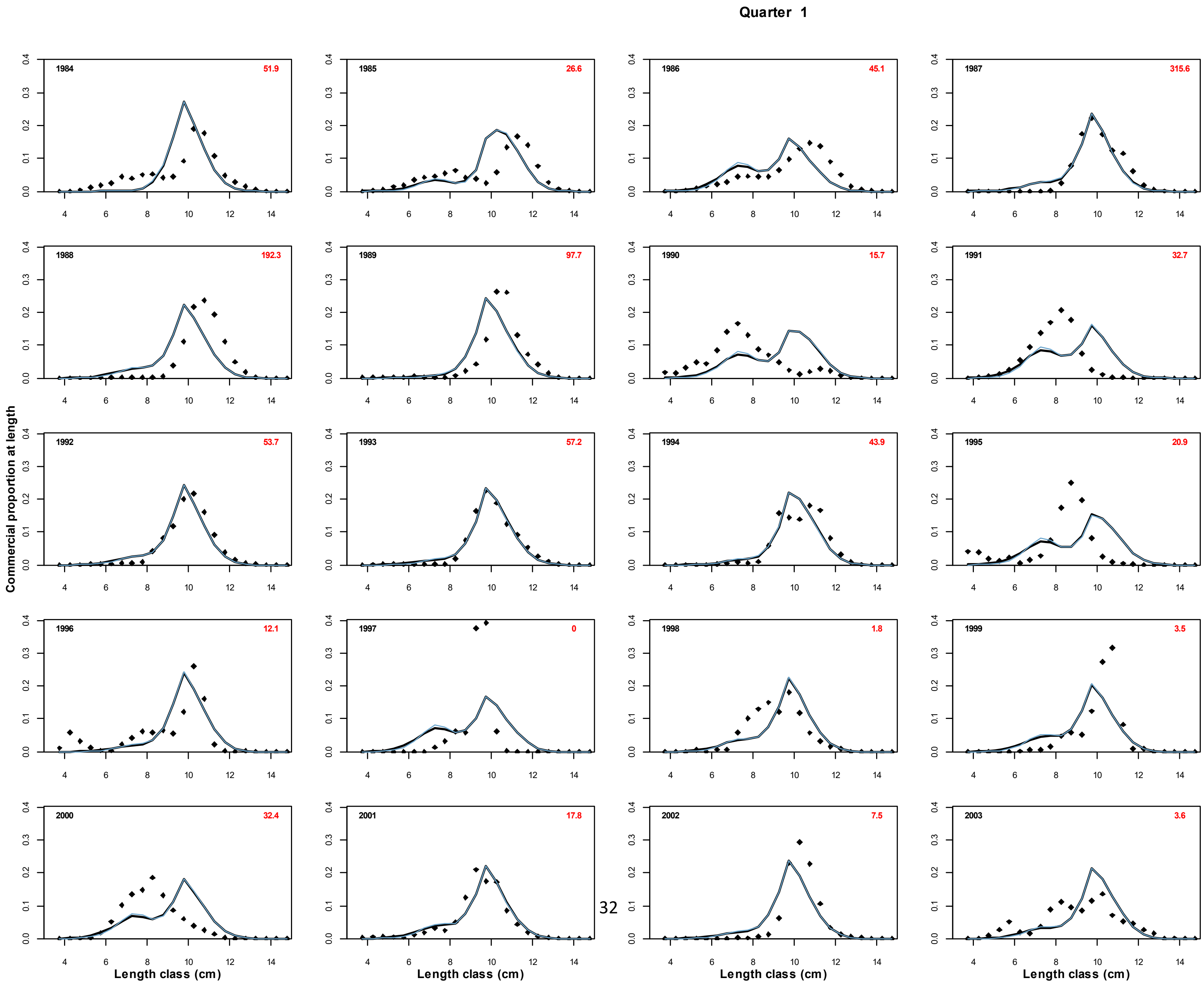
[23] This is the observed length-frequency adjusted such that the expected mass calculated using the weight-at-length relationship matches the observed catch in tons (de Moor *et al*. 2024).
[24] No data available. Set equal to 1985.
[25] No data available. Used average 2020, 2022.
[26] No data for November 2023 available. Used average 2017-2020, 2022.

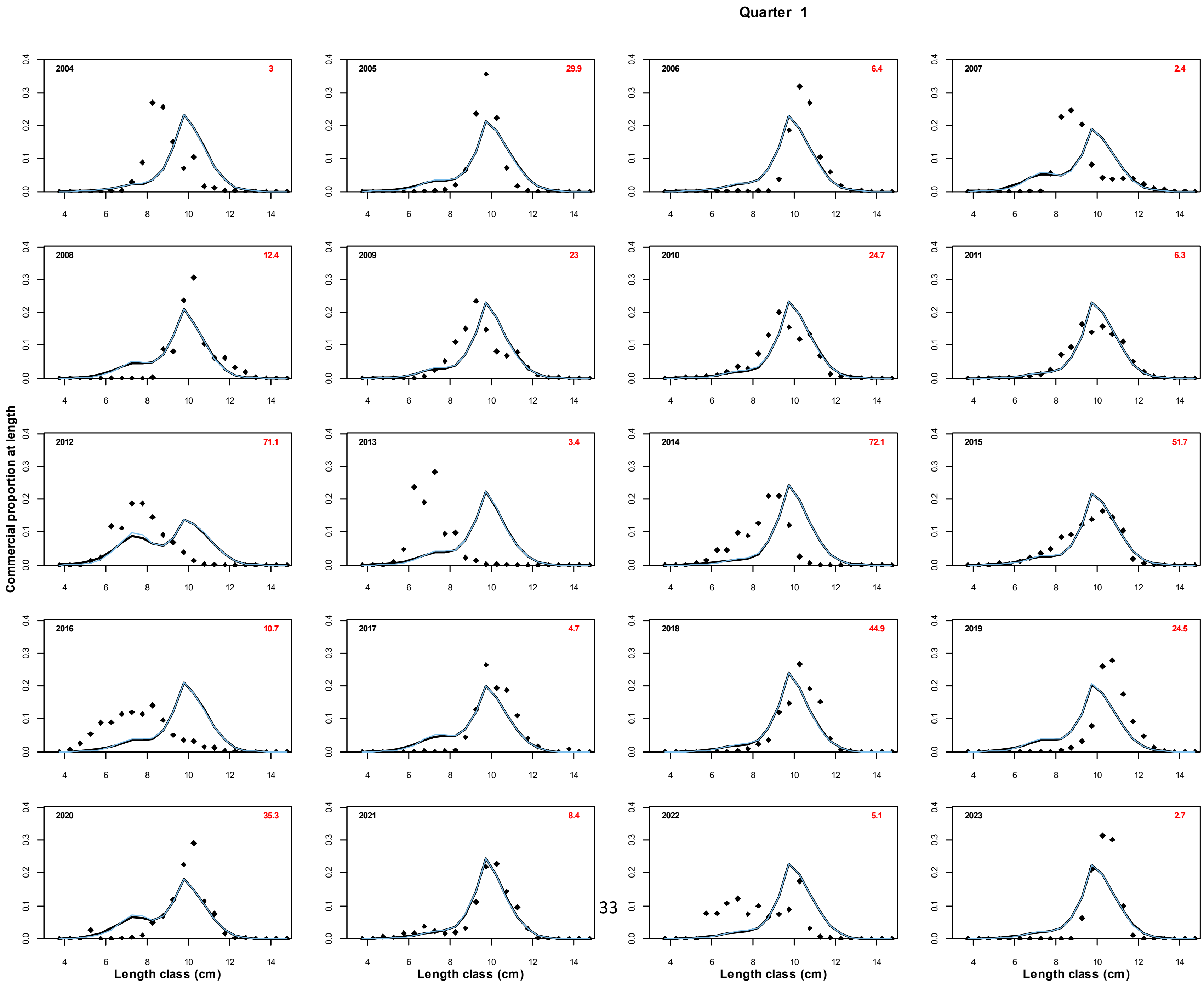
Appendix B: "Hardly informative" prior distributions

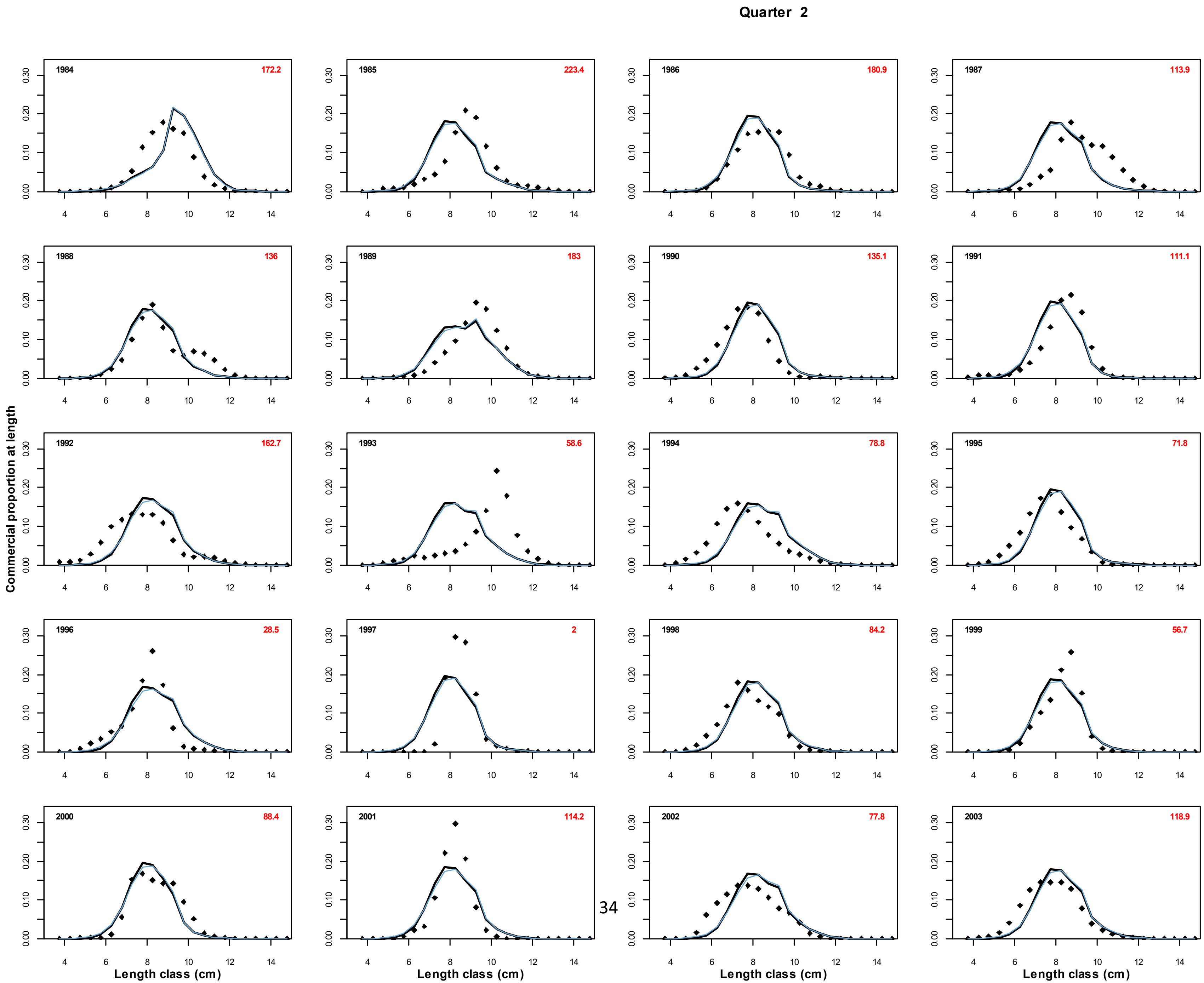
Based on difficulties with convergence to the posterior distribution from previous assessments (de Moor 2016), "hardly informative" prior distributions have been used for some parameters which do no more than simply aid the software to conduct MCMC simulations.

Normal prior distributions were assigned to these parameters with means roughly corresponding to the baseline parameter values giving the minimum objective function value. The standard deviations for these prior distributions were chosen such that the Hessian-based SE resulting from the model fit was less (as much less as possible) than that of the prior distribution.

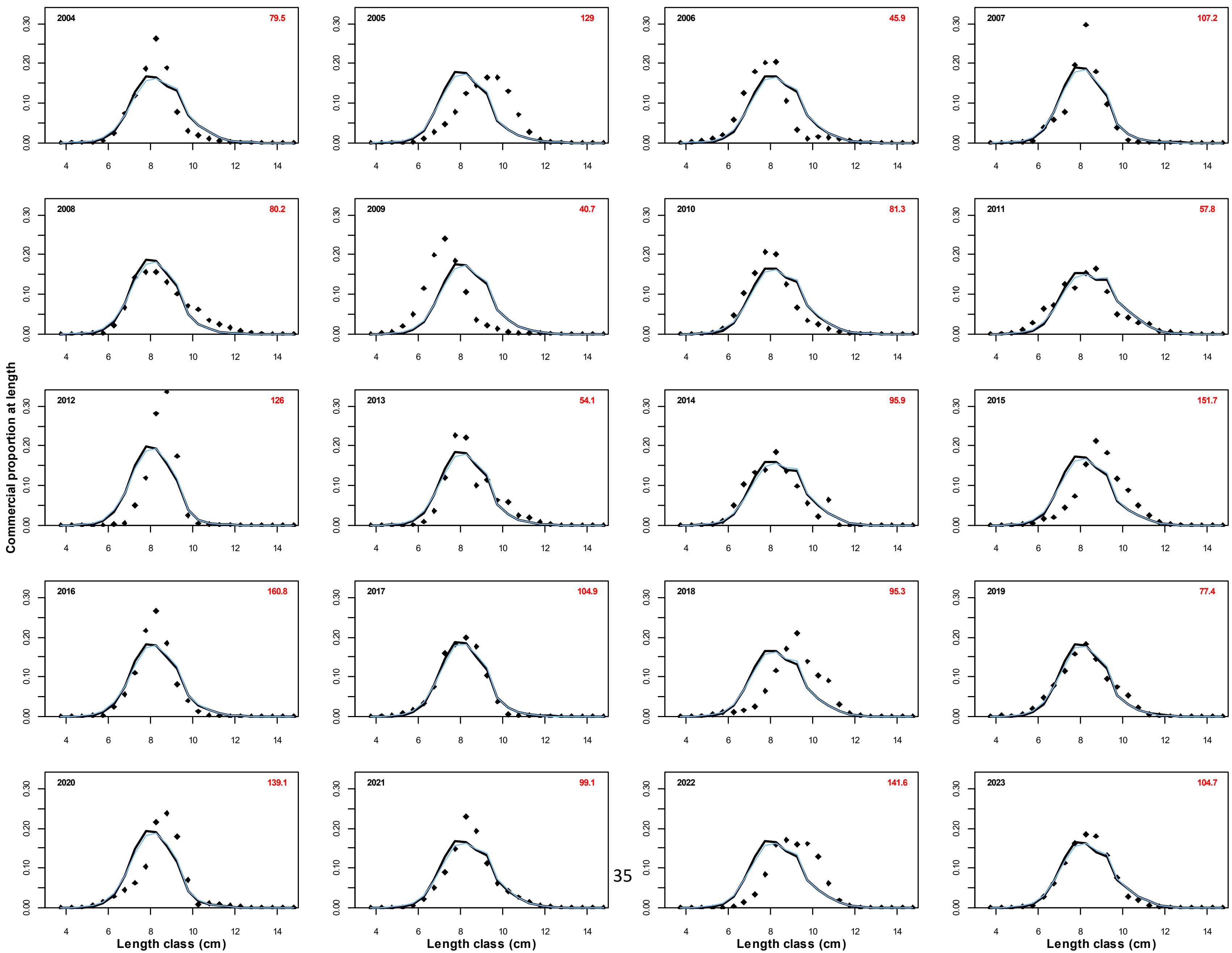
Appendix C: Model predicted and observed proportions-at-length in the quarterly commercial catch for A_{BH} (black) and a model where all quarterly commercial selectivity parameters are estimated (blue).

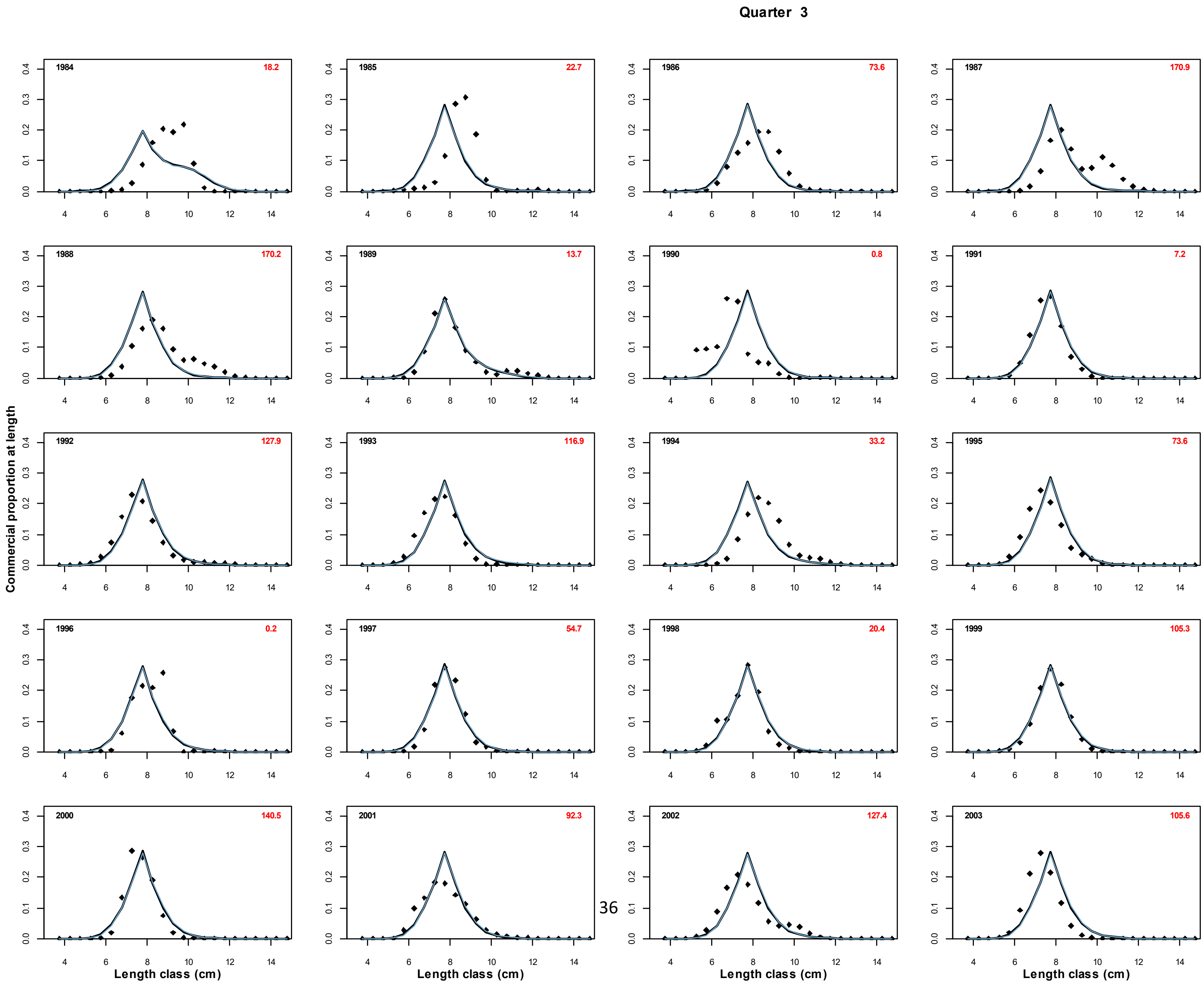

Quarter 1
Commercial proportion at length
Length class (cm)
1984 51.9
1985 26.6
1986 45.1
1987 315.6
1988 192.3
1989 97.7
1990 15.7
1991 32.7
1992 53.7
1993 57.2
1994 43.9
1995 20.9
1996 12.1
1997 0
1998 1.8
1999 3.5
2000 32.4
2001 17.8
2002 7.5
2003 3.6

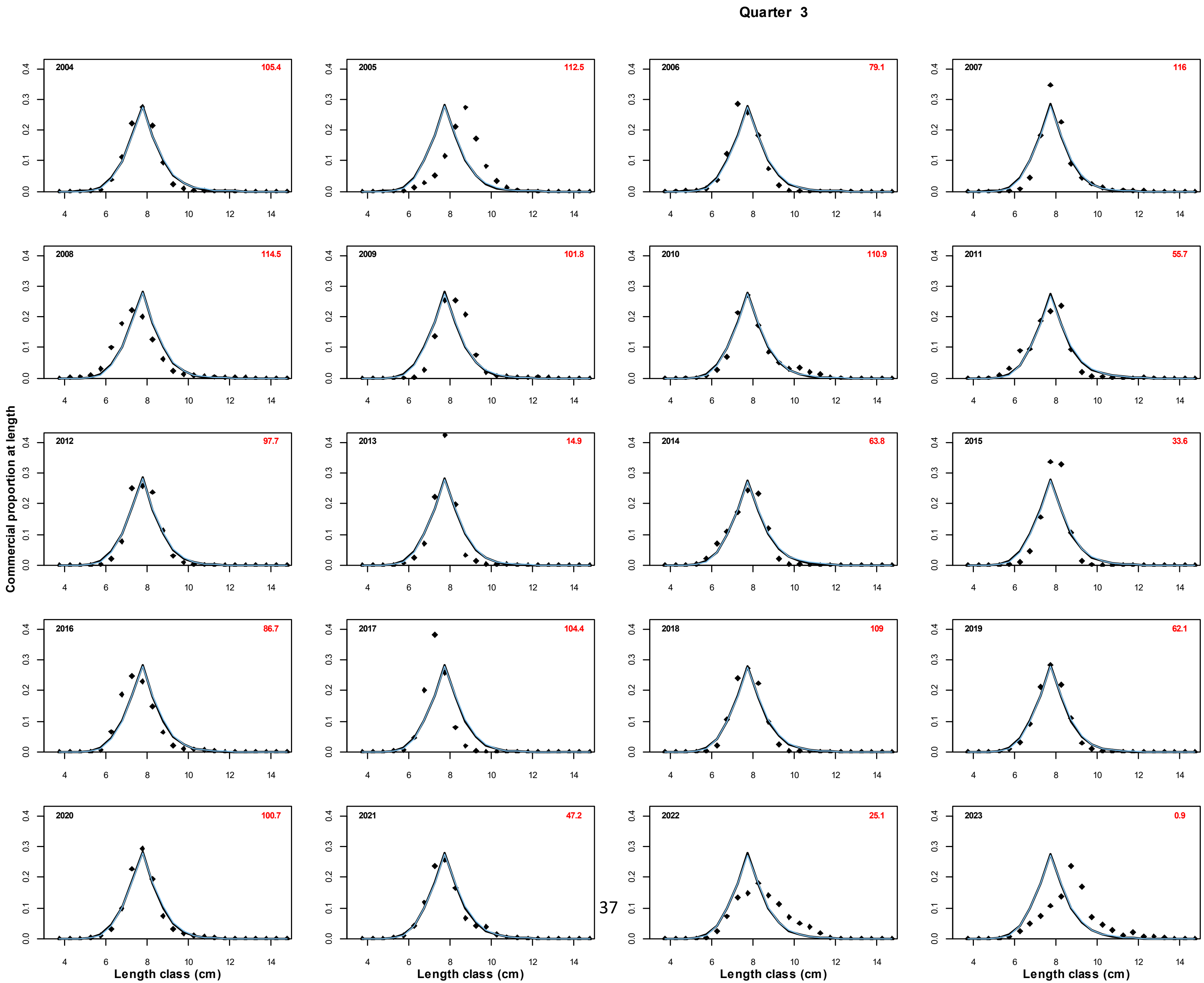

Quarter 1
Commercial proportion at length
Length class (cm)
2004 3
2005 29.9
2006 6.4
2007 2.4
2008 12.4
2009 23
2010 24.7
2011 6.3
2012 71.1
2013 3.4
2014 72.1
2015 51.7
2016 10.7
2017 4.7
2018 44.9
2019 24.5
2020 35.3
2021 8.4
2022 5.1
2023 2.7

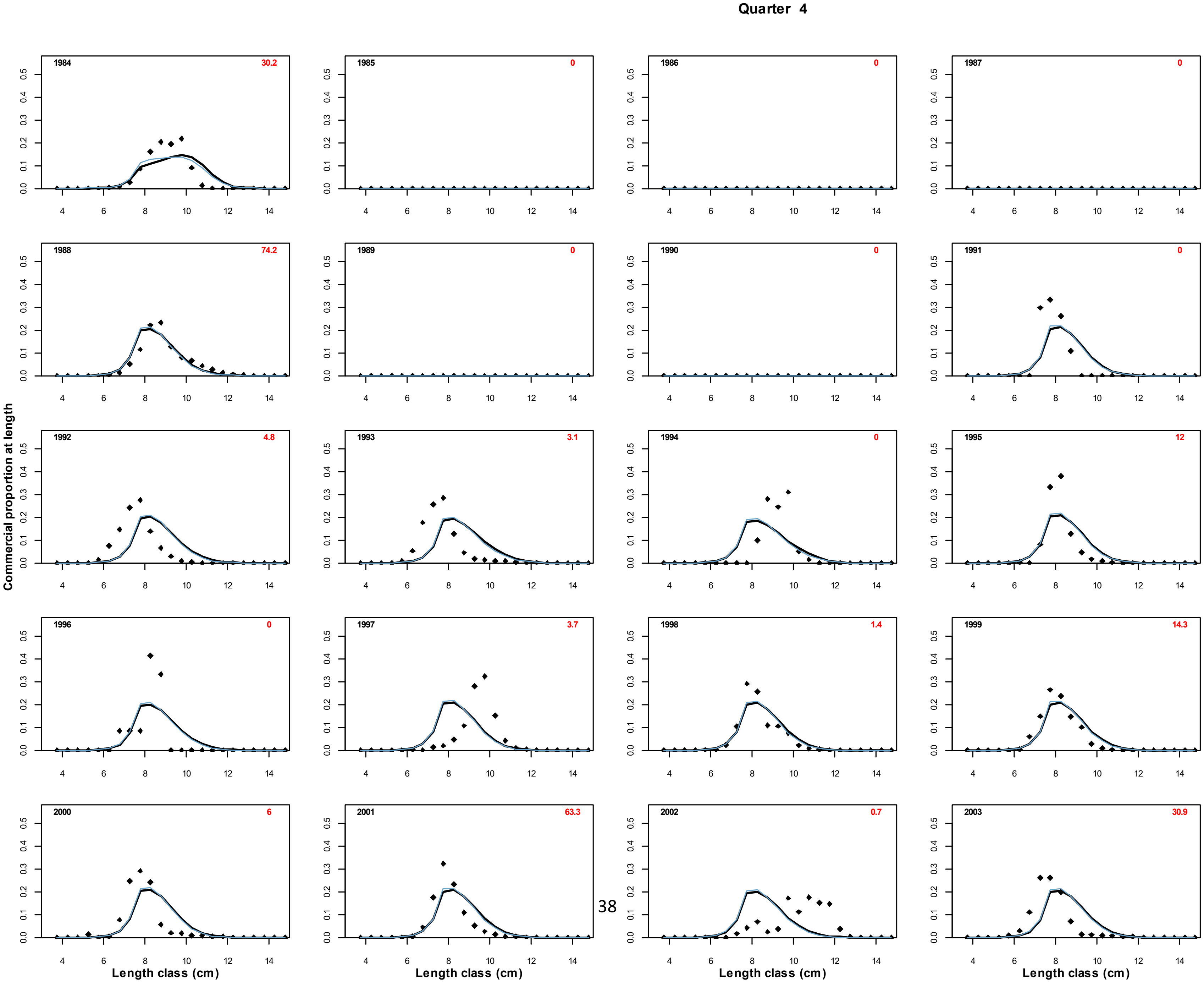

Quarter 2
Commercial proportion at length
Length class (cm)
1984
172.2
1985
223.4
1986
180.9
1987
113.9
1988
136
1989
183
1990
135.1
1991
111.1
1992
162.7
1993
58.6
1994
78.8
1995
71.8
1996
28.5
1997
2
1998
84.2
1999
56.7
2000
88.4
2001
114.2
2002
77.8
2003
118.9

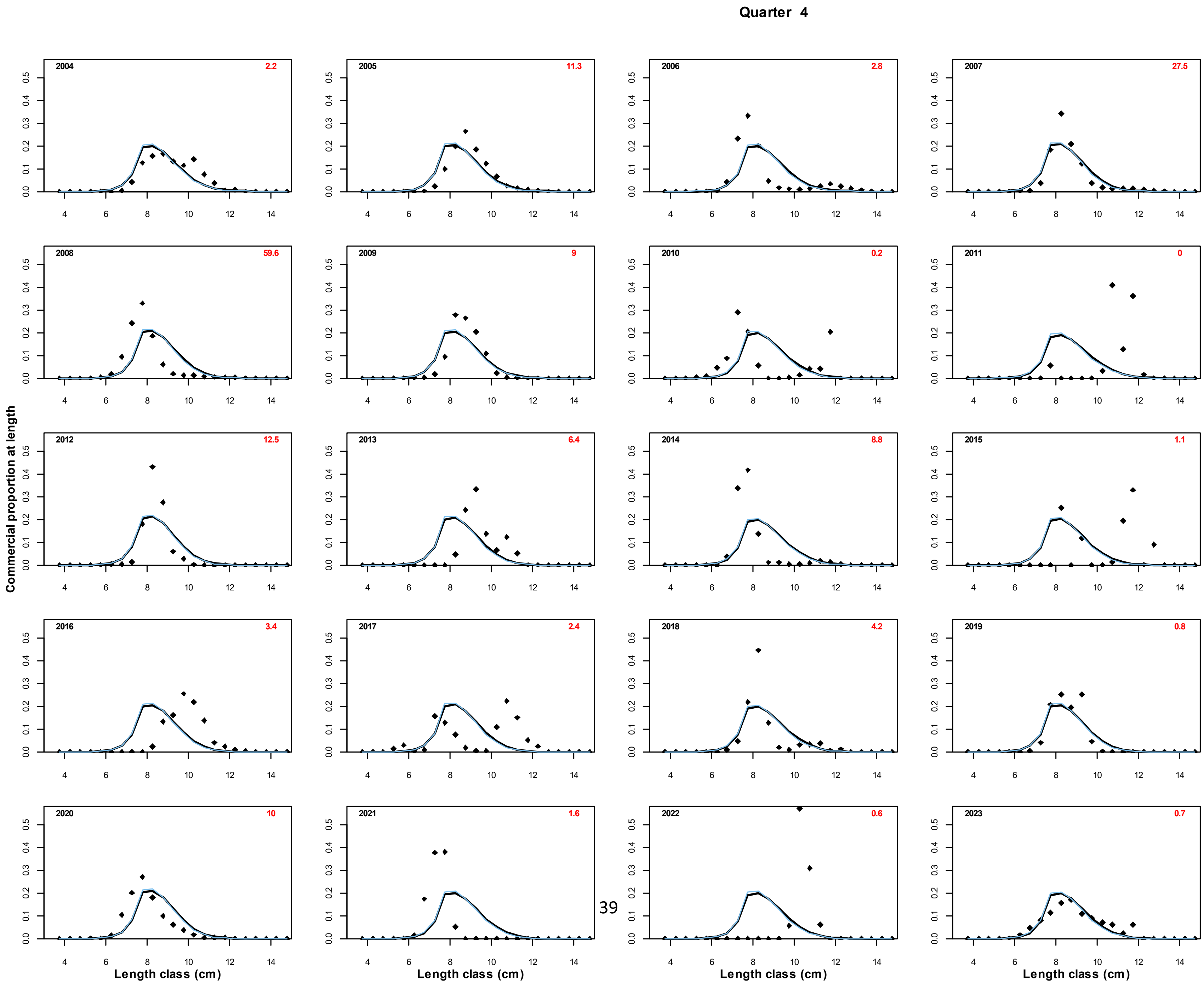
Quarter 2


2004
79.5
2005
129
2006
45.9
2007
107.2
2008
80.2
2009
40.7
2010
81.3
2011
57.8
2012
126
2013
54.1
2014
95.9
2015
151.7
2016
160.8
2017
104.9
2018
95.3
2019
77.4
2020
139.1
2021
99.1
2022
141.6
2023
104.7
Commercial proportion at length
Length class (cm)
0.00
0.10
0.20
0.30
4
6
8
10
12
14


Quarter 3
Commercial proportion at length
Length class (cm)
1984 18.2
1985 22.7
1986 73.6
1987 170.9
1988 170.2
1989 13.7
1990 0.8
1991 7.2
1992 127.9
1993 116.9
1994 33.2
1995 73.6
1996 0.2
1997 54.7
1998 20.4
1999 105.3
2000 140.5
2001 92.3
2002 127.4
2003 105.6


Quarter 3
Commercial proportion at length
Length class (cm)
2004 105.4
2005 112.5
2006 79.1
2007 116
2008 114.5
2009 101.8
2010 110.9
2011 55.7
2012 97.7
2013 14.9
2014 63.8
2015 33.6
2016 86.7
2017 104.4
2018 109
2019 62.1
2020 100.7
2021 47.2
2022 25.1
2023 0.9


Quarter 4
Commercial proportion at length
Length class (cm)
1984
30.2
1985
0
1986
0
1987
0
1988
74.2
1989
0
1990
0
1991
0
1992
4.8
1993
3.1
1994
0
1995
12
1996
0
1997
3.7
1998
1.4
1999
14.3
2000
6
2001
63.3
2002
0.7
2003
30.9


Quarter 4
Commercial proportion at length
Length class (cm)
2004 2.2
2005 11.3
2006 2.8
2007 27.5
2008 59.6
2009 9
2010 0.2
2011 0
2012 12.5
2013 6.4
2014 8.8
2015 1.1
2016 3.4
2017 2.4
2018 4.2
2019 0.8
2020 10
2021 1.6
2022 0.6
2023 0.7